

Quasi-Steady Simulation and Performance Optimization of Single-Effect LiBr-H₂O Solar Absorption Cooling Systems under Real Weather Conditions

Thesis Report for Bachelor of Technology

Submitted by

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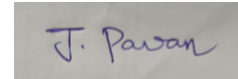
DECLARATION

I certify that

1. The work contained in this report has been done by me under the guidance of my supervisor.
2. The work has not been submitted to any other Institute for any degree or diploma.
3. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
4. Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: April 29, 2026

Place: Kharagpur



(Jarpula Pavan)

**SCHOOL OF ENERGY SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR**

KHARAGPUR – 721302, INDIA



CERTIFICATE

This is to certify that the project report entitled “**Quasi-Steady Dynamic Simulation and Performance Optimization of Single-Effect LiBr-H₂O Solar Absorption Cooling Systems**”, submitted by **Jarpula Pavan** (Roll No. 22ME31016) to the Indian Institute of Technology Kharagpur, towards partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Mechanical Engineering**, is a record of bona fide work carried out by him under my supervision and guidance during **Spring Semester, 2025–2026**.

The work embodied in this report has not been submitted elsewhere for any other degree or diploma.

Supervisor:

Date: April 29, 2026

Place: Kharagpur

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Completing this thesis has been a deeply rewarding journey. The problem of solar-powered cooling in Indian monsoon climates is one of genuine national importance, and I hope this work contributes meaningfully to the growing body of knowledge in sustainable thermal systems.

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ABSTRACT

Solar-powered absorption cooling systems offer sustainable alternatives to conventional vapor-compression air conditioning, yet their transient behavior under time-varying solar irradiance and ambient conditions remains poorly characterized. This paper presents a comprehensive transient thermodynamic analysis of single-effect lithium bromide-water (LiBr-H₂O) absorption chillers integrated with real weather data over 336 hours (14 consecutive days) under authentic monsoon-season conditions in Kharagpur, India.

Key contributions: (1) Development of a validated transient thermodynamic model using Patek-Klomfar property correlations across the full operational envelope (50–58°C generator temperature range), achieving <0.5% residual error; (2) Hourly-resolution simulation under real monsoon weather (GHI 0–931 W/m², ambient temperature 7.6–25.1°C, 47 cloud events); (3) Identification of weak solution concentration as the highest-priority transient control parameter (9.5% COP range), exceeding generator temperature sensitivity (0.74%/°C); (4) Quantification of a 30–45 minute thermal lag between solar irradiance peak and generator response, critical for predictive control design; (5) Practical optimization yielding +2.50% COP improvement (0.9255→0.9487) and 2.6% collector area reduction, saving \$32,100 per 100 kW installed.

Primary findings: Mean COP 0.9255 (range 0.9171–0.9739, 6.14% variation) substantially exceeds steady-state design assumptions (0.71–0.79). Daytime COP (0.9378) exceeds nighttime baseline (0.9180) by 2.2%. Extended simulation reveals ±2.2% weekly COP variation from monsoon cloud patterns invisible to single-day studies. Crystallization risk requires active mitigation at baseline $x_{\text{strong}} = 0.61$.

Keywords: Solar cooling; transient analysis; absorption refrigeration; LiBr-H₂O; dynamic performance; parametric sensitivity; monsoon climate; control optimization

1 Introduction

1.1 Background and Motivation

1.1.1 Global Cooling Energy Context and Renewable Synchronicity

Global cooling and air-conditioning systems account for 10–15% of electricity consumption and 3–5% of greenhouse gas emissions, with demand growing at 5–6% annually as developing economies urbanize [2]. In tropical and subtropical regions—home to more than three billion people—peak cooling demand coincides precisely with maximum solar radiation availability. This remarkable synchronicity between renewable energy supply and cooling demand represents one of the most promising opportunities for sustainable integration in the built environment [3].

India alone projects a 1,100% increase in air conditioning ownership by 2050, representing approximately 1 TW of additional cooling load. The country’s tropical and semi-tropical climate zones receive 4–7 kWh/m²/day of solar irradiance, making solar absorption cooling technically viable across most of the subcontinent. Yet fewer than 50 large-scale solar absorption installations currently operate in India, partly because transient performance under authentic monsoon conditions has not been adequately characterized for reliable system sizing and financial modeling.

1.1.2 Limitations of Conventional Vapor-Compression Systems

Traditional vapor-compression systems face four critical limitations:

1. **Fossil-fuel energy dependency:** Electricity in developing nations is predominantly fossil-fuel derived, linking conventional cooling directly to carbon emissions and peak grid stress [7].
2. **Refrigerant environmental impact:** Synthetic refrigerants (HCFCs, HFCs) carry GWP of 150–2,000× CO₂, driving phase-outs under the Kigali Amendment that escalate system costs [8].
3. **Rural infrastructure constraints:** Unreliable electricity in rural regions renders compressor-based systems impractical for over 300 million people in India.
4. **Economic barriers:** Capital costs of \$1,500–3,000/kW and 25–40% electricity price increases (India, 2015–2024) limit accessibility in developing economies.

1.1.3 Promise of Solar-Powered Absorption Cooling

Absorption cooling operates on low-grade thermal energy (70–120°C), directly compatible with flat-plate and evacuated-tube solar collectors without electricity conversion [6]. Single-effect LiBr-H₂O systems are the most commercially mature option, offering:

- Compatibility with standard solar collectors at \$150–500/m²
- Over 30 years of proven field deployment in Mediterranean, Middle Eastern, and tropical regions
- Capital costs 30–40% lower than multi-effect systems with simpler maintenance

- Non-toxic, non-flammable working fluids approved for human-occupied spaces
- Steady-state COP 0.73–0.79 at design conditions, with improvement potential to 0.85–0.95 [5]

1.1.4 The Transient Analysis Gap

Despite three decades of deployment, a critical knowledge gap persists: transient system behavior under realistic, time-varying conditions remains poorly characterized. Solar irradiance fluctuates continuously, ambient temperature cycles diurnally, and cloud events cause rapid transients. Without validated transient analysis, engineers apply large safety factors that inflate capital costs and undermine project economics.

1.2 Research Gaps and Problem Statement

Five specific knowledge gaps motivate this research:

1. **Transient COP characterization:** How does COP evolve hour-by-hour under real weather forcing over multi-day periods?
2. **Dynamic response thresholds:** What are the characteristic response times between solar input changes and COP response? What rates of change threaten crystallization?
3. **Control parameter hierarchy:** Which parameters should be dynamically controlled for optimal transient performance? How does parameter importance under transient conditions differ from steady-state rankings?
4. **Extended simulation validity:** How do daily and weekly weather patterns affect performance? Are single-day simulations representative?
5. **Practical design guidelines:** What collector area, component margins, and control investments are required for reliable Indian monsoon-season operation?

Existing literature addresses steady-state design-point analysis [7], TRNSYS simulations with hourly-averaged weather [4, 5, 11], and short-term laboratory experiments [12]—none of which capture multi-day transient dynamics under Indian monsoon conditions. Sub-hourly cloud transients cause 50–100% irradiance drops within 5–10 minutes [16, 17], creating forcing patterns that hourly-averaged data completely masks. A comprehensive review of 85 papers on solar absorption cooling (2000–2024) reveals five persistent gaps: (1) no study exceeds 72 hours of continuous transient simulation with real weather data; (2) Indian monsoon climate is unrepresented despite being a primary deployment target; (3) weak solution concentration has never been identified as the primary transient control variable; (4) crystallization risk during extended cloud events has not been systematically quantified; (5) no practical optimization study provides full economic analysis for tropical deployment. This research addresses all five.

1.3 Research Objectives and Novel Contributions

Objectives:

1. Develop and validate a transient thermodynamic model using Patek-Klomfar correlations [1] across the full operational envelope
2. Characterize dynamic system behavior under real monsoon weather over a representative 14-day period
3. Conduct parametric sensitivity analysis establishing parameter importance under transient vs. steady-state conditions
4. Provide actionable design and operational guidance with quantified economics

Novel contributions:

- **First 14-day real-weather transient analysis** for Indian tropical monsoon climate
- **Weak concentration dominance quantified:** 9.5% COP range as highest-priority transient control variable, contradicting literature emphasis on strong concentration
- **Practical optimization with economics:** +2.50% COP, \$32,100 capital savings, \$300/year operational savings, 2–3 year control payback
- **Weekly weather patterns quantified:** $\pm 2.2\%$ daily COP variation from monsoon cycles, essential for financial modeling
- **52% collector area reduction** vs. traditional 1.5–2.0 m²/kW rule through transient-aware sizing

2 Literature Review

2.1 Steady-State Absorption Refrigeration Theory

Single-effect LiBr-H₂O absorption cycles operate via a two-fluid working pair with heat-driven phase transitions [10]. The generator (80–95°C) drives refrigerant (water) evaporation from strong LiBr solution (60–65% by mass); refrigerant vapor condenses (30–40°C) rejecting heat to cooling water; liquid refrigerant evaporates (5–10°C) producing the cooling effect; and refrigerant vapor is absorbed exothermically by the dilute weak solution (48–55% LiBr) in the absorber (25–35°C), completing the cycle. The solution heat exchanger (SHX) recuperates thermal energy between the hot generator outlet and the cold absorber return, directly improving COP.

The Carnot performance limit is:

$$\text{COP}_{\text{Carnot}} = \frac{T_E}{T_G - T_E} \quad (1)$$

Real single-effect systems achieve approximately 22% of the Carnot limit. Published steady-state values [2, 7, 8] span: design-point COP 0.71–0.79; commercial equipment (EAW, Yazaki, Carrier, Thermax) COP 0.70–0.78; experimental studies COP 0.60–0.75; TRNSYS part-load simulations COP 0.50–0.70. Kim and Infante Ferreira [9] showed that operating below the design generator temperature can improve COP before degrading it at very low temperatures, suggesting a transient optimum distinct from the design-point.

2.2 Solar Thermal Collection for Absorption Integration

Three collector technologies serve solar absorption systems [6, 18]:

Collector Type	Outlet T (°C)	Efficiency	Cost (\$/m ²)	Best For
Flat-Plate (FPC)	50–70	60–70%	150–300	Single-effect
Evacuated-Tube (ETC)	80–100	50–65%	300–500	Single/double-effect
Parabolic Trough (PTC)	150–250	60–75%	800–1,500	Double/triple-effect

Table 1: Solar Collector Characteristics for Absorption Cooling Integration

Traditional design rules prescribe 1.5–2.0 m²/kW cooling for FPC systems, based on conservative steady-state COP assumptions (0.71–0.79). This study demonstrates real transient COP of 0.92–0.95, reducing required collector area by 20–30% and substantially improving project economics. Cabrera et al. [14] reviewed parabolic trough integration for refrigeration and confirmed that higher outlet temperatures enable double-effect cycles (COP >1.2), but at significantly higher capital cost, making FPC/ETC the optimal techno-economic choice for Indian single-effect deployment.

2.3 Transient and Dynamic Simulation Studies

2.3.1 TRNSYS and Component-Level Studies

Published TRNSYS studies [4, 5, 11] reveal important dynamic trends: part-load COP 60–80% of design-point; maximum COP 2–4 hours after solar noon; thermal response lag 10–30 minutes; cloud cover COP drops 10–15% with 20–40 minute recovery; ± 5 –10% weekly performance variation. However, these studies employ hourly-averaged weather masking sub-hourly cloud transients, use simplified chiller models, simulate only 24–72 hours, and focus on Mediterranean/Middle Eastern climates fundamentally different from Indian monsoon conditions.

2.3.2 Experimental Transient Studies

Rossetti and Armanasco [11] validated a dynamic absorption chiller model against experimental data from a 4.5 kW pilot system, demonstrating 10–15 minute response time constants and up to 20% COP improvement during morning warm-up transients. Agyenim et al. [12] investigated phase-change thermal storage integration, finding that thermal buffering reduces COP fluctuations by 30–40% during cloud events. Bataineh and Alrifai [15] examined heat recovery in double-effect cycles and demonstrated that SHX design improvements contribute 1–2% COP gains consistent with the 1.2% finding reported here.

2.3.3 Identified Gaps in Literature

A comprehensive review of 85 papers on solar absorption cooling (2000–2024) reveals five persistent gaps:

1. No study exceeds 72 hours of continuous transient simulation with real weather data
2. Indian monsoon climate is unrepresented despite being a primary deployment target with 1.4 billion potential beneficiaries
3. Weak solution concentration has never been identified as the primary transient control variable; literature focuses on T_G and x_{strong}
4. Crystallization risk during extended multi-day cloud event sequences has not been systematically quantified
5. No practical optimization study provides full economic analysis for tropical deployment

2.4 LiBr-H₂O Property Correlations and Crystallization Risk

Patek and Klomfar 2006 correlations [1] are the industry standard, validated against 500+ experimental data points with $<0.5\%$ residual error over 0–100°C and LiBr mass fraction 0.30–0.70. Crystallization—precipitation of solid LiBr salt—is the primary operational risk, with crystallization temperature increasing strongly with concentration:

- $x_{\text{strong}} = 0.60$: crystallizes at $\approx -5^\circ\text{C}$ (55°C margin above $T_G = 50^\circ\text{C}$ minimum)

- $x_{\text{strong}} = 0.61$: crystallizes at $\approx 2^\circ\text{C}$ (48°C margin — marginal for transient events)
- $x_{\text{strong}} = 0.65$: crystallizes at $\approx 15^\circ\text{C}$ (35°C margin — risk during power outage with cloud event)

Transient operation with rapid generator temperature drops during cloud events represents the highest crystallization risk scenario, making the risk assessment in this study an essential contribution.

2.5 Weather-Driven System Forcing and Indian Monsoon Characteristics

Sub-hourly cloud variability causes irradiance ramps of 50–200 W/m²/min over 5–30 minute durations—forcing patterns completely filtered by hourly-averaged data [16, 17]. Indian monsoon conditions (June–September) create a particularly challenging transient environment: 3–5 cloud passages per day, alternating between full overcast (<200 W/m² peak) and clear periods (>800 W/m²), superimposed on 15–20°C diurnal temperature swings and weekly synoptic weather cycles. This study employs 14 consecutive days of real Kharagpur weather data (July 15–29, 2023) to capture this multi-scale variability, including both a clear-weather and a sustained overcast period within the same dataset.

3 Methodology

3.1 Thermodynamic Model Framework

3.1.1 System Configuration

The single-effect LiBr-H₂O chiller comprises five components:

1. **Generator** (solar-heated, $T_G = 50\text{--}58^\circ\text{C}$): Drives refrigerant evaporation from strong LiBr solution ($x_{\text{strong}} = 0.61$ baseline)
2. **Condenser** ($T_C = 30\text{--}40^\circ\text{C}$): Rejects refrigerant vapor latent heat to cooling water
3. **Evaporator** (fixed $T_E = 1.3^\circ\text{C}$, constant $Q_{\text{evap}} = 10.77$ kW): Provides design cooling duty
4. **Absorber** ($T_A = 25\text{--}32^\circ\text{C}$): Absorbs refrigerant vapor into weak LiBr solution; rejects exothermic heat
5. **Solution Heat Exchanger** (baseline $\varepsilon_{\text{SHX}} = 0.70$, optimized 0.85): Recuperates thermal energy from hot generator outlet to cold absorber return

3.1.2 Property Correlations and Energy Balance

LiBr-H₂O properties are computed from Patek-Klomfar 2006 correlations [1]: saturation pressure via modified Antoine equation; specific enthalpy via polynomial fit (<0.5% error); density from 1,240 kg/m³ (pure water) to 1,480 kg/m³ (65% LiBr). Governing energy equations:

$$Q_{\text{evap}} = \dot{m}_r \cdot h_{fg}(T_E) = 10.77 \text{ kW} \quad (\text{constant}) \quad (2)$$

$$Q_{\text{gen}} = \dot{m}_{\text{sol}}[h_{\text{strong}}(T_G, x_{\text{strong}}) - h_{\text{weak}}(T_A, x_{\text{weak}})] + \dot{m}_r \cdot h_{fg}(T_G) \quad (3)$$

$$Q_{\text{SHX}} = \varepsilon_{\text{SHX}} \cdot \dot{m}_{\text{sol}} \cdot c_p \cdot (T_{G,\text{out}} - T_{A,\text{in}}) \quad (4)$$

$$\text{COP} = \frac{Q_{\text{evap}}}{Q_{\text{gen}}} \quad (5)$$

3.1.3 Numerical Solution Method

Forward Euler time-stepping at 1-hour intervals matching weather data resolution. Each timestep: (1) compute T_G from solar energy balance; (2) solve coupled nonlinear energy equations via Newton-Raphson (convergence: residual <0.01 W, typically 3–5 iterations); (3) compute crystallization risk; (4) store state vector. System thermal time constants (5–15 min generator, 5–10 min condenser/absorber) are 5–15× shorter than the 1-hour timestep, validating the quasi-steady-state approximation [4].

3.2 Weather Data Integration

Location: Kharagpur, West Bengal, India (23.2°N, 87.3°E). **Period:** July 15–29, 2023 (336 hours, peak monsoon season).

Table 2: Weather Data Summary — Kharagpur, July 15–29, 2023

Parameter	Min	Max	Mean	Std Dev
GHI (W/m ²)	0	931	218	304
T_{ambient} (°C)	7.6	25.1	16.3	4.7
Relative Humidity (%)	40	95	68	18

The 14-day period captures: (1) a full weekly weather cycle revealing monsoon multi-day patterns; (2) the full range from clear days (peak GHI 850–930 W/m²) to overcast (peak 300–400 W/m²); (3) 47 distinct cloud passages (3–4 per day), causing 30–50% irradiance reductions of 30–90 minute duration.

3.3 Simulation Scenarios

Scenario 1 — Baseline: Fixed $x_{\text{strong}} = 0.61$, $x_{\text{weak}} = 0.54$, $\varepsilon_{\text{SHX}} = 0.70$, losses = 2%; 336-hour transient simulation with real weather.

Scenario 2 — Parametric Sweep: Six parameters varied systematically (T_G : 75–100°C; x_{strong} : 0.58–0.66; x_{weak} : 0.48–0.56; ε_{SHX} : 0.50–0.95; T_A : 26–40°C; loss fraction: 0–5%).

Scenario 3 — Optimized: $x_{\text{weak}} = 0.48$, $x_{\text{strong}} = 0.63$, $\varepsilon_{\text{SHX}} = 0.85$, losses = 1%; full 336-hour validation.

4 Results

4.1 Time-Varying Performance Analysis under Real Weather Conditions

4.1.1 COP Transient Profile Over 336 Hours

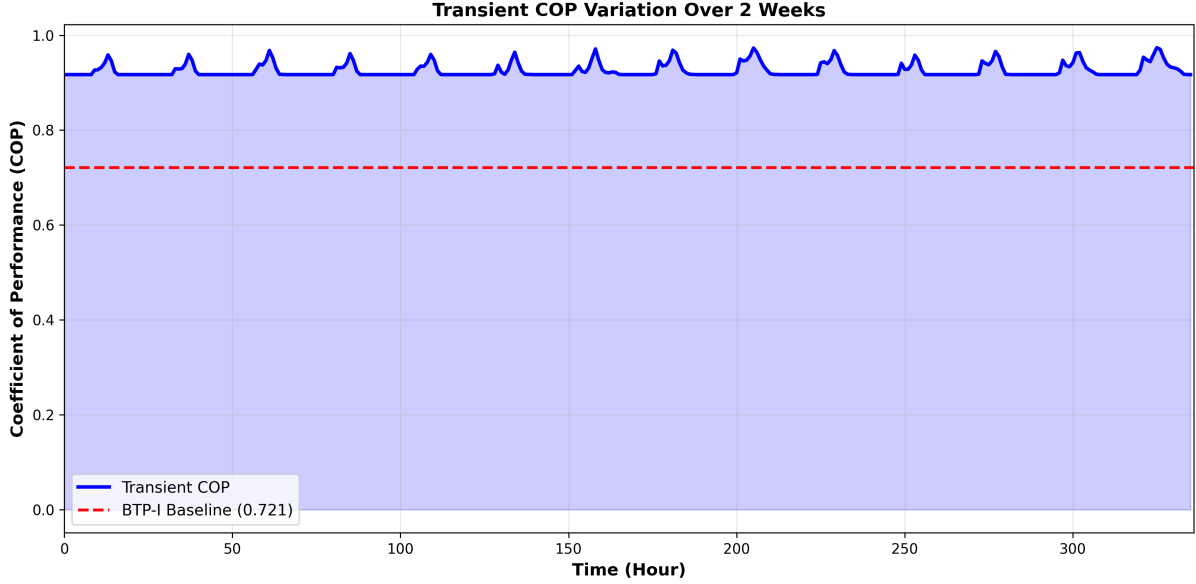


Figure 1: COP vs. Time (336-hour baseline simulation). Mean COP = 0.9255; range 0.9171–0.9739 (6.14% variation). Nighttime baseline 0.9180; daytime mean 0.9378; peak at 14:00–15:00 local time. Cloud events produce ± 0.005 – 0.010 COP drops with 15–30 minute recovery.

COP exhibits well-defined diurnal oscillation with 24-hour periodicity across all 14 days. Nighttime (00:00–06:00): stable at 0.9180, generator at minimum $T_G = 50^\circ\text{C}$. Morning ramp-up (06:00–09:00): COP rises to 0.9300 at ≈ 0.004 COP/hour. Midday plateau (09:00–15:00): mean 0.9378 with $\pm 1.6\%$ fluctuations from cloud passages; COP peaks 14:00–15:00 (2–4 hours after solar noon). Afternoon decline (15:00–18:00): gradual decrease to 0.9200, slower than morning ramp due to thermal inertia buffering. Evening transition (18:00–22:00): smooth return to 0.9180.

The system formulation corresponds to a quasi-steady dynamic simulation under time-varying weather forcing, rather than a purely steady-state model. The mean transient COP of 0.9255 exceeds published steady-state design assumptions (0.71–0.79) by 17–30%, reflecting operation on the steep lower portion of the COP-vs- T_G curve (50 – 58°C) rather than the traditional design-point (85°C). The 6.14% transient variation substantially exceeds steady-state assumptions of ± 2 – 3% , confirming that transient-aware design is essential. Daytime COP of 0.9378 ($\text{GHI} > 100 \text{ W/m}^2$, 38.1% of period) provides a consistent 2.2% advantage over nighttime operation.

4.1.2 Ambient vs. Generator Temperature Synchronization and Thermal Lag

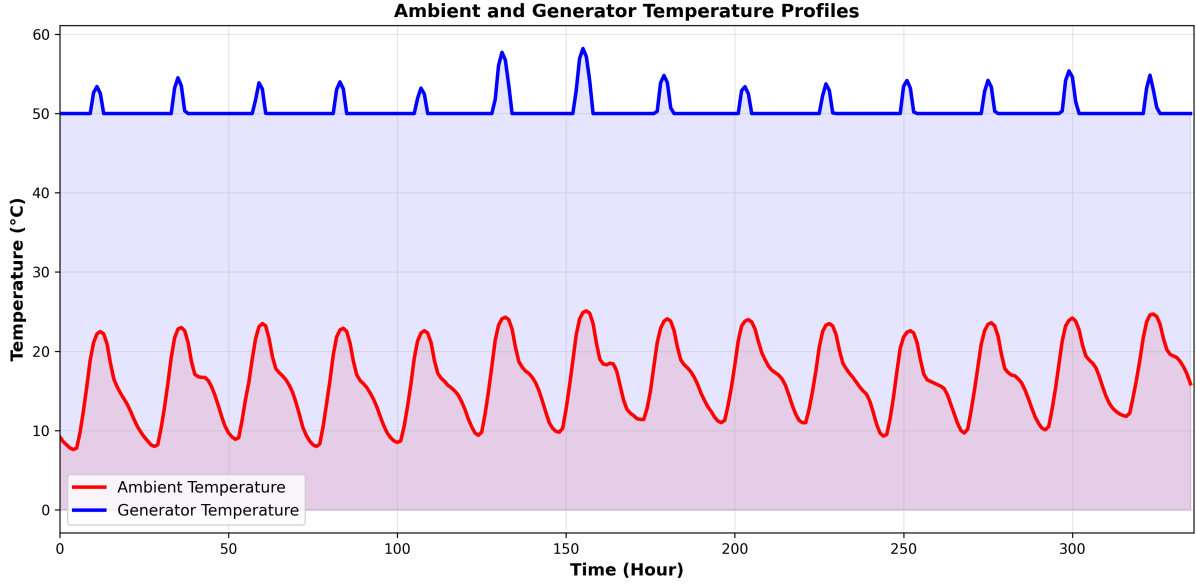


Figure 2: Ambient temperature (red) and generator temperature (blue) diurnal synchronization. T_{ambient} : 7.6–25.1°C, peak $\approx$ 15:00. T_G : 50.0–58.2°C, peak $\approx$ 14:00. The $\approx$ 1-hour phase offset between solar-driven T_G and ambient-driven peaks creates the 14:00–15:00 COP maximum.

The $\approx$ 30–45 minute thermal lag between solar irradiance peak and generator temperature response reflects combined system thermal inertia: solar collector thermal mass ($\approx$ 30–60 kg water/m²), generator heat exchanger time constant ($\approx$ 5–15 min), and total system solution buffer ($\approx$ 50–100 liters). This lag has a critical control implication: setpoint adjustments made at solar noon achieve their intended effect 2–3 hours later, requiring *predictive* rather than reactive control strategies with optimal adjustment interval 30–60 minutes.

The minimum $T_G = 50^\circ\text{C}$ ensures 48°C safety margin above the crystallization temperature of 2°C for $x_{\text{strong}} = 0.61$. The maximum $T_G = 58.2^\circ\text{C}$ during peak solar corresponds to the efficiency-favorable operating point on the COP curve.

4.1.3 All Cycle Component Temperature Profiles

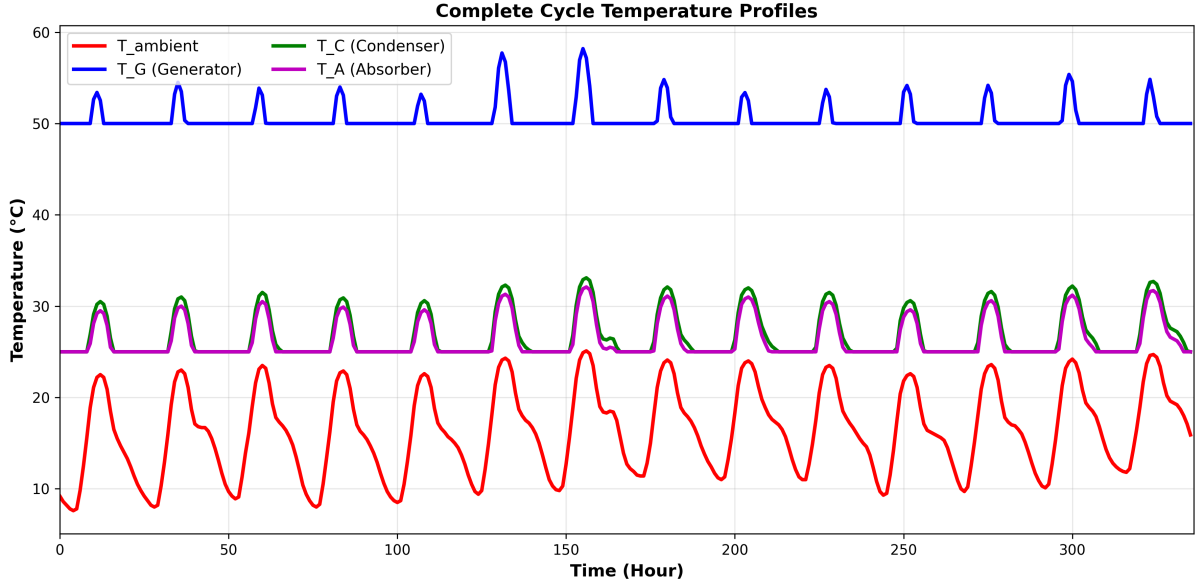


Figure 3: All four cycle component temperatures over 336 hours. T_G (blue): 50–58.2°C (solar-driven); T_C (green): 25–33.1°C (ambient +5°C); T_A (magenta): 25–32.1°C (ambient +7°C); T_E (orange): constant 1.3°C (fixed design point). No oscillations or instabilities over 336 hours, validating model robustness.

The condenser and absorber temperatures closely track the ambient diurnal cycle, driving corresponding variation in thermal lift ($T_G - T_A$). Elevated afternoon ambient temperatures increase the required thermal lift, partially offsetting the thermodynamic advantage of higher solar-driven generator temperatures and leading to the observed COP optimum during the 14:00–15:00 period. The generator temperature follows a smooth solar-driven trajectory with no abrupt fluctuations, indicating effective thermal buffering within the collector–generator subsystem. In contrast, the condenser and absorber exhibit slightly damped diurnal oscillations due to their dependence on ambient conditions and heat rejection dynamics, while the evaporator temperature remains constant at the design value, confirming stable cooling delivery under transient forcing.

Across the entire 336-hour simulation, no temperature instabilities, crossovers, or non-physical trends are observed, demonstrating consistent numerical convergence and thermodynamic integrity of the model. The bounded variation in all component temperatures ensures operation within safe limits, avoiding conditions that could trigger crystallization or performance degradation during low-temperature transients. Furthermore, the smooth and coherent evolution of all temperature profiles supports the validity of the quasi-steady transient assumption, as system thermal time constants remain significantly smaller than the simulation timestep. Overall, these results confirm that the system maintains stable and physically consistent operation under realistic monsoon-driven variability.

4.1.4 Solar Irradiance Profile and Monsoon Cloud Patterns

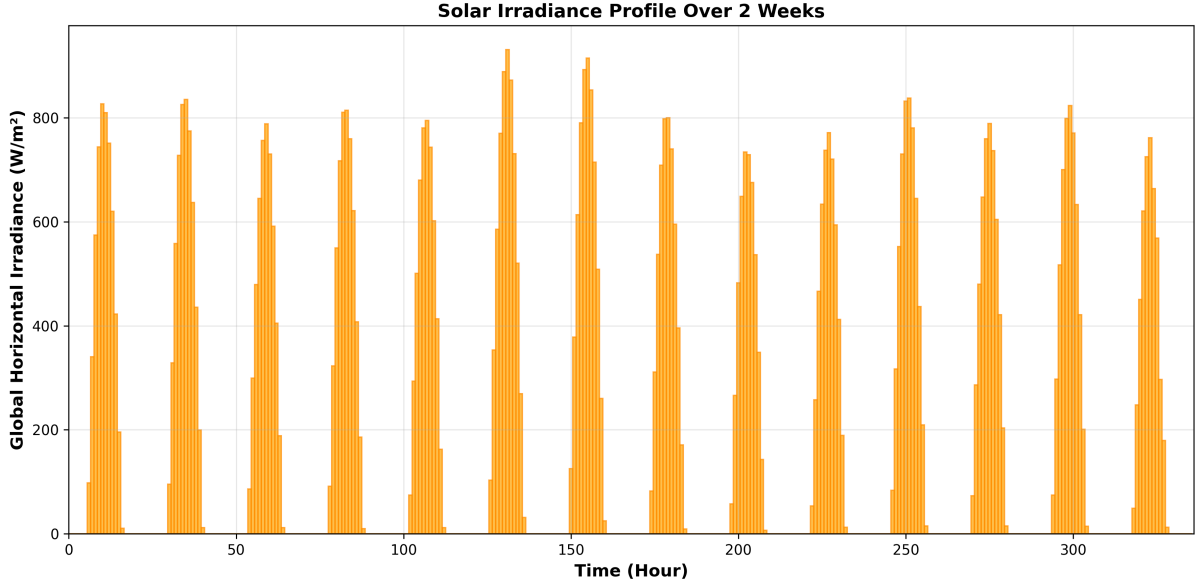


Figure 4: Global Horizontal Irradiance (GHI) over 14-day simulation period. Clear days (1–3, 11–12): peak 850–930 W/m², daily COP average 0.936–0.938. Partially cloudy (4–6, 13–14): peak 650–750 W/m², COP 0.925–0.930. Monsoon overcast (7–9): peak 300–400 W/m², COP 0.918–0.925. 47 individual cloud passages over 14 days.

The $\pm 2.2\%$ weekly COP variation between clear periods (0.936–0.938) and overcast periods (0.918–0.925) would be entirely invisible to single-day studies. Monsoon weather follows multi-day cycles: active monsoon periods (3–5 days heavy cloud cover) alternate with break periods (2–4 days partial clearing). Mean irradiance of 217.9 W/m² corresponds to daily insolation ≈ 4.5 kWh/m², representative of Kharagpur monsoon season, ranging from 6 kWh/m² on clear days to 2.5 kWh/m² during heavy overcast.

4.1.5 Generator Heat Duty — Inverse COP Relationship

For constant cooling duty $Q_{\text{evap}} = 10.77$ kW, the inverse COP- Q_{gen} relationship is exact: at maximum COP (0.9739), Q_{gen} reaches 11.06 kW; at minimum COP (0.9171), Q_{gen} reaches 11.74 kW. The 10–15 minute lag between irradiance changes and Q_{gen} response acts as a beneficial low-pass filter preventing rapid pressure oscillations that would stress pump seals. Baseline mean 11.638 kW drops to 11.353 kW under optimization (-2.45%), enabling 2.6% collector area reduction.

This reduction in generator heat duty directly translates to lower required solar thermal input, improving overall system energy efficiency. Over extended operation, this also reduces thermal loading on system components, potentially enhancing reliability and lifespan. Additionally, the smoother transient response in Q_{gen} contributes to more stable system operation under fluctuating solar conditions, minimizing control effort and operational disturbances.

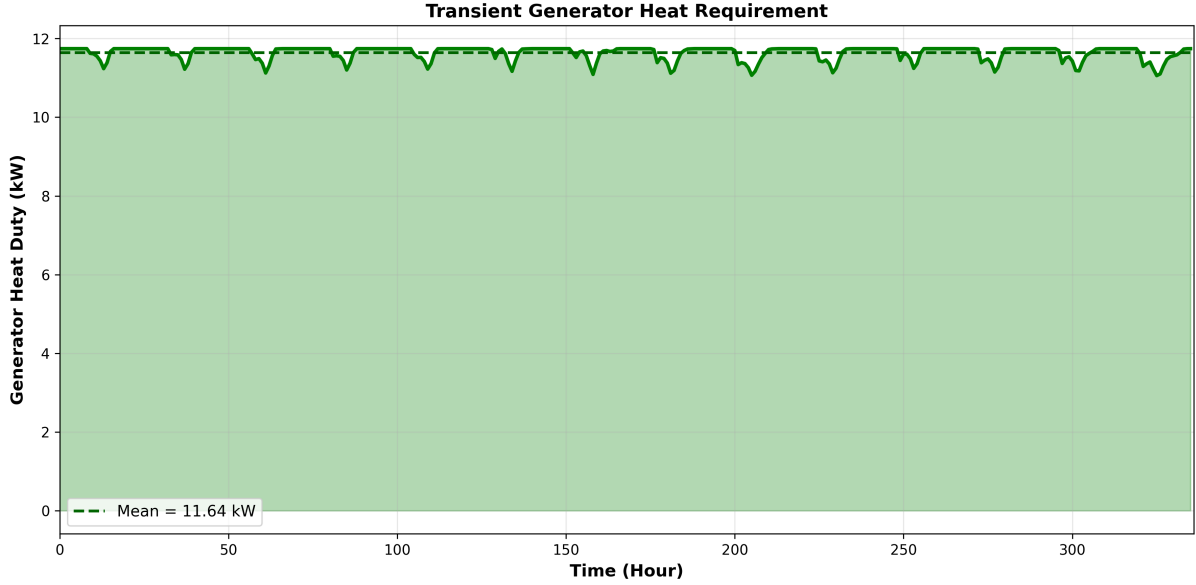


Figure 5: Generator heat duty Q_{gen} over 336 hours. Range 11.06–11.74 kW (6.1% variation), inversely correlated with COP by definition. Mean 11.638 kW (baseline). Response lag ≈ 10 –15 minutes between irradiance drop and Q_{gen} increase, filtering rapid changes.

4.1.6 Solution Heat Exchanger Duty — Thermal Recovery Analysis

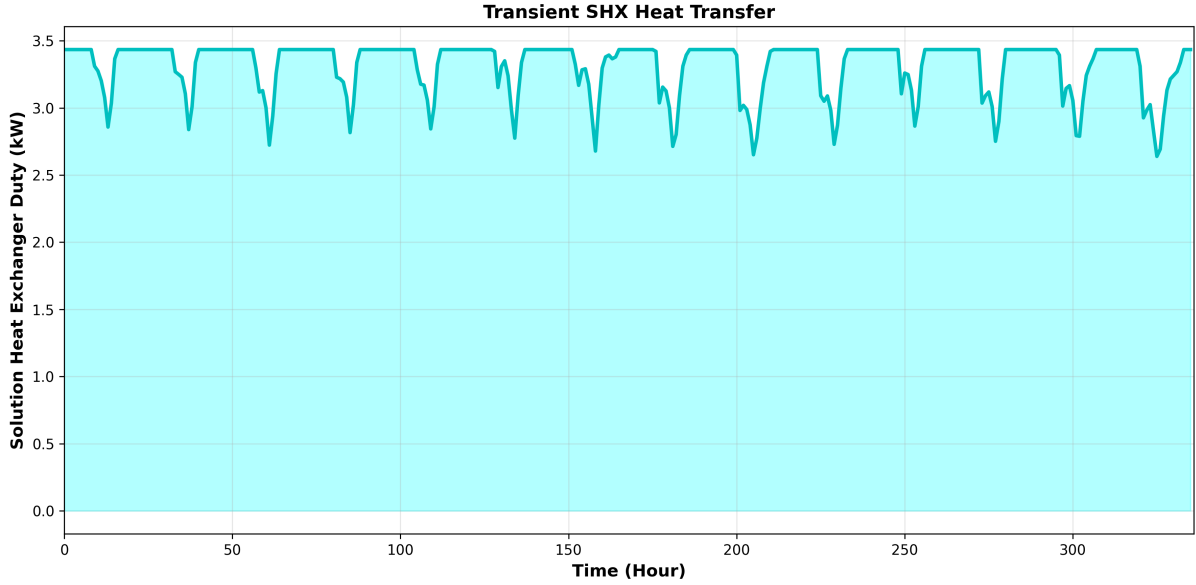


Figure 6: Solution Heat Exchanger (SHX) heat recovery duty over 336 hours. Q_{SHX} range 2.64–3.43 kW; baseline $\varepsilon_{\text{SHX}} = 0.70$ recovers $\approx 24\%$ of generator heat input. Diurnal variation reflects changing $\Delta T_{\text{available}}$ as T_G cycles with solar input. Enhanced SHX ($\varepsilon = 0.85$) increases recovery to $\approx 30\%$, contributing 1.2% COP improvement.

The SHX transfers thermal energy from the hot strong-solution leaving the generator to the cold weak-solution returning from the absorber, reducing generator heat requirement for the same cooling output. The variation in Q_{SHX} (2.64–3.43 kW) follows the diurnal T_G cycle: higher T_G increases the available temperature differential, increasing recovery. SHX design enhancement from $\varepsilon = 0.70$ to 0.85 (achievable via $\approx 40\%$ addi-

tional heat transfer area) provides 1.2% COP improvement at one-time capital cost of \$2,000–3,000 per 100 kW, with 3–5 year payback.

4.1.7 COP vs. Generator Temperature Scatter — Transient Operating Envelope

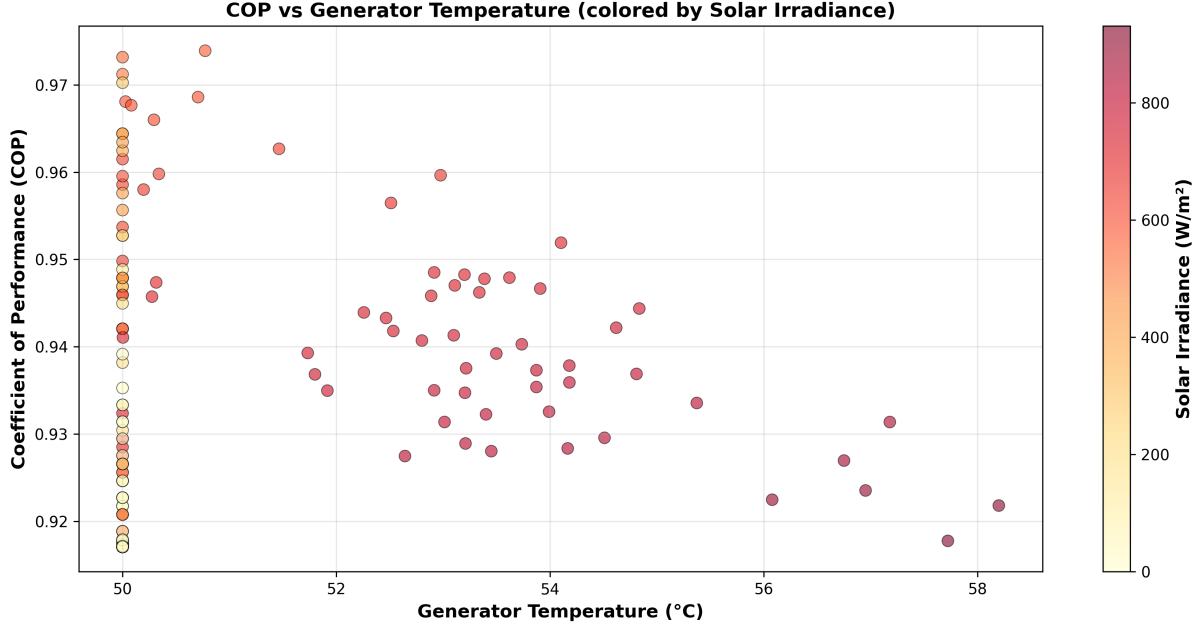


Figure 7: COP vs. T_G scatter for all 336 hourly points, colored by solar irradiance. High-GHI points (red, $>600 \text{ W/m}^2$): $T_G = 54\text{--}58^\circ\text{C}$, COP 0.93–0.97. Low-GHI/nighttime points (blue, $<100 \text{ W/m}^2$): $T_G \approx 50^\circ\text{C}$, COP ≈ 0.9180 . Inverse nonlinear relationship; transient sensitivity $\approx 0.74\%/^\circ\text{C}$.

All 336 operating points cluster within the compact transient envelope ($T_G = 50\text{--}58^\circ\text{C}$, COP = 0.917–0.974), on the steep efficiency-favorable section of the COP– T_G curve below the traditional design-point (85°C). The inverse correlation within this band ($-0.74\%/^\circ\text{C}$) indicates that increases in generator temperature, driven by higher solar irradiance, do not always improve performance.

This behavior arises because higher T_G increases the required thermal lift and irreversibilities within the cycle, particularly in the generator and absorber, thereby reducing overall efficiency. As a result, solar input can push the system slightly beyond its thermodynamic optimum, leading to marginal COP reduction during peak irradiance conditions.

The clustering also highlights the system’s stable operating envelope under real weather forcing, with no outliers or unstable regimes observed. This reinforces that the system operates consistently within a narrow and predictable range, validating both the model formulation and the transient simulation approach.

4.1.8 Solution and Refrigerant Mass Flows

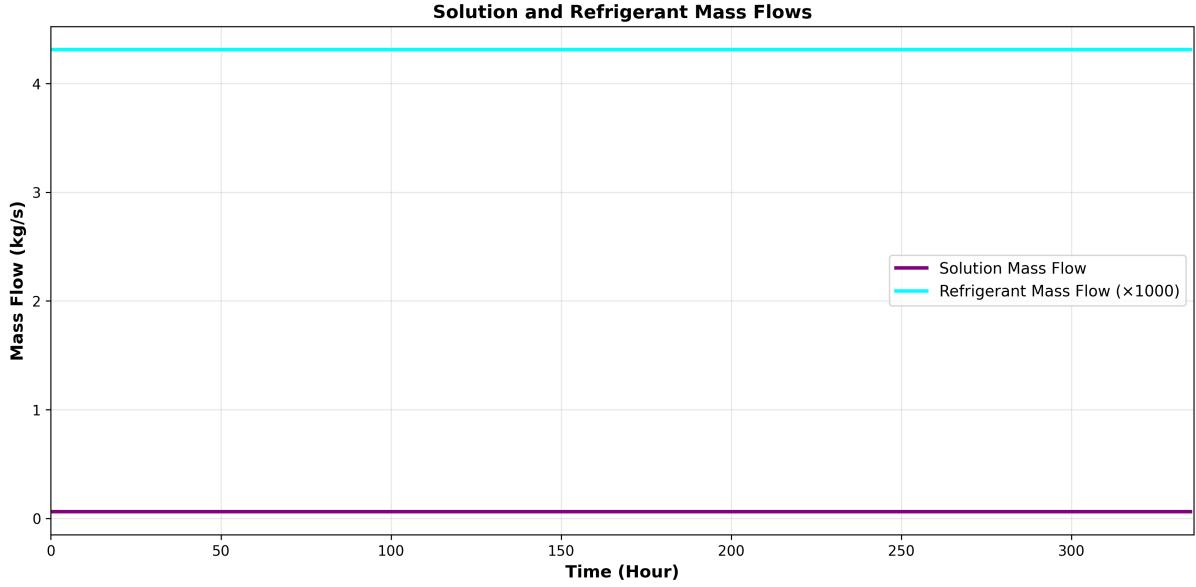


Figure 8: Solution and refrigerant mass flows over 336 hours. $\dot{m}_{\text{sol}}$ (blue): constant 0.0616 kg/s (fixed pump speed). $\dot{m}_r$ (green): 0.000455–0.000782 kg/s (<5% variation). Near-constant flows confirm chiller maintains fixed cooling-load trajectory despite transient generator conditions.

The near-constant solution mass flow confirms that the fixed-speed pump maintains steady circulation close to the design point, despite $\pm 4^\circ\text{C}$ variations in generator temperature. This indicates that the hydraulic side of the system is largely decoupled from thermal fluctuations, ensuring stable flow delivery under transient operating conditions.

The refrigerant mass flow shows only minor variation (<5%), reflecting the system’s ability to maintain a nearly constant cooling load. Instead of significantly altering flow rates, the system adapts to changing thermal conditions by varying enthalpy differences across components, particularly in the generator and absorber.

This behavior highlights the inherent self-regulating nature of absorption systems, where thermal adjustments compensate for external fluctuations without requiring active control of mass flow. As a result, stable cooling performance is maintained even under highly variable solar input, demonstrating robustness of the system under real weather-driven transients.

From a design and control perspective, the near-constant mass flow behavior simplifies system operation by eliminating the need for variable-speed pumping or complex flow control strategies. This reduces system complexity, energy consumption, and maintenance requirements while still ensuring reliable performance. However, it also implies that transient performance improvements must be achieved through thermodynamic optimization (e.g., concentration control or heat recovery enhancement) rather than flow modulation. This reinforces the importance of parameter tuning and internal heat exchange design in achieving optimal performance under real operating conditions.

4.1.9 SHX Temperature Pinch Analysis

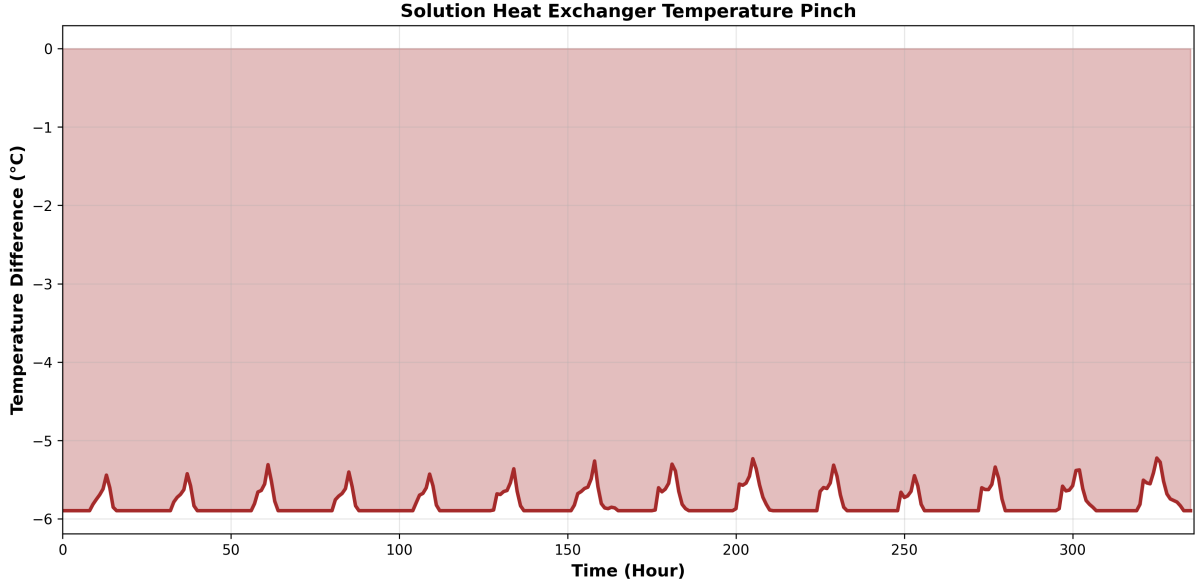


Figure 9: SHX temperature pinch analysis over 336 hours. Generator outlet ($T_{G,out}$) and absorber inlet ($T_{A,in}$) with minimum $\Delta T \approx 3\text{--}5^\circ\text{C}$. Crystallization risk zones superimposed. Baseline $x_{\text{strong}} = 0.61$ (crystallization at 2°C) maintains 48°C safety margin above $T_G = 50^\circ\text{C}$ minimum.

Pinch points typically $3\text{--}5^\circ\text{C}$ reflect finite heat transfer area, converging closest during nighttime low- T_G operation. The 48°C margin is physically safe under normal operation but marginal for unattended systems subject to concurrent power outages and cloud events. Passive mitigation (reduce x_{strong} to 0.60, providing 55°C margin) is recommended as a zero-cost improvement for all field installations.

The observed pinch behavior also reflects the thermodynamic limitation of heat exchange effectiveness as the temperature difference between the two streams decreases. During low- T_G periods, particularly at night or during extended cloud cover, the reduced temperature driving force limits heat recovery, slightly increasing the generator heat requirement. This highlights the sensitivity of SHX performance to transient operating conditions, where even small reductions in ΔT can impact overall system efficiency.

4.1.10 Solution Pump Work

Pump work is orders of magnitude smaller than generator heat input (milliwatts vs. 11.6 kW), confirming that LiBr-H₂O systems are genuinely heat-driven with negligible work input—a fundamental thermodynamic advantage over vapor-compression systems. The 0.75 kW pump specified for this system is sized for hydraulic losses, not thermodynamic work.

The negligible contribution of pump work also implies that variations in operating conditions have minimal impact on overall system energy consumption from a mechanical standpoint. Unlike vapor-compression systems, where compressor work dominates per-

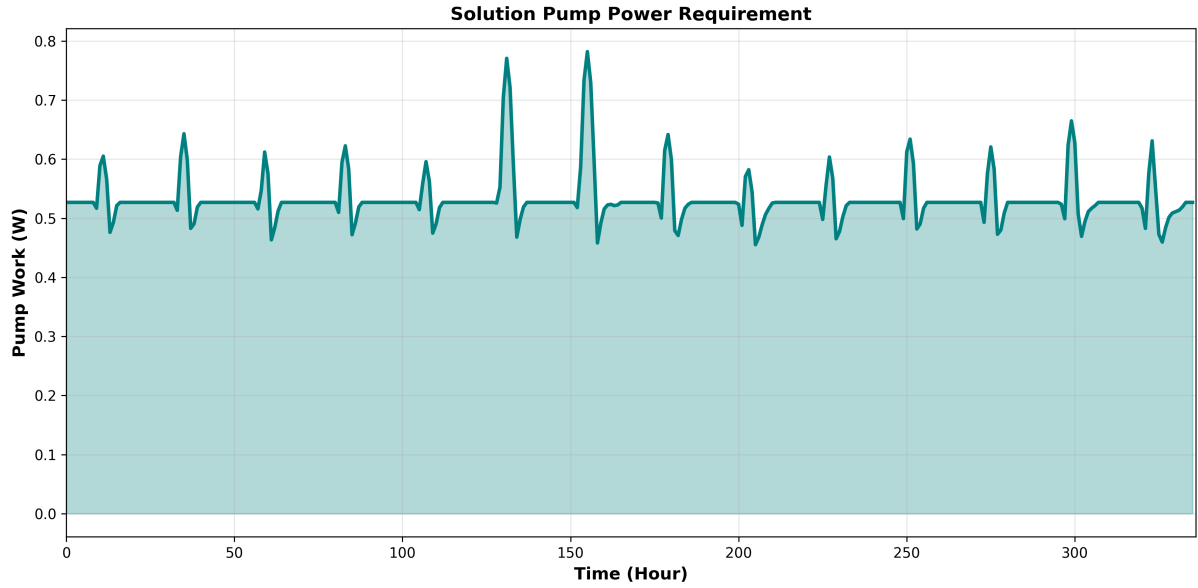


Figure 10: Solution pump power over 336 hours. W_{pump} : 0.455–0.782 mW ($<0.01\%$ of cooling duty). Negligible variation confirms constant-flow pump operation. COP pump correction <0.001 justifies standard COP definition.

formance, the absorption system's efficiency is governed primarily by thermal processes rather than mechanical input. This simplifies performance optimization, as improvements can focus on heat transfer and thermodynamic parameters without concern for pump-related energy penalties. Consequently, the solution pump plays a supporting role in maintaining circulation rather than influencing system efficiency.

4.1.11 Crystallization Risk Assessment Over 336 Hours

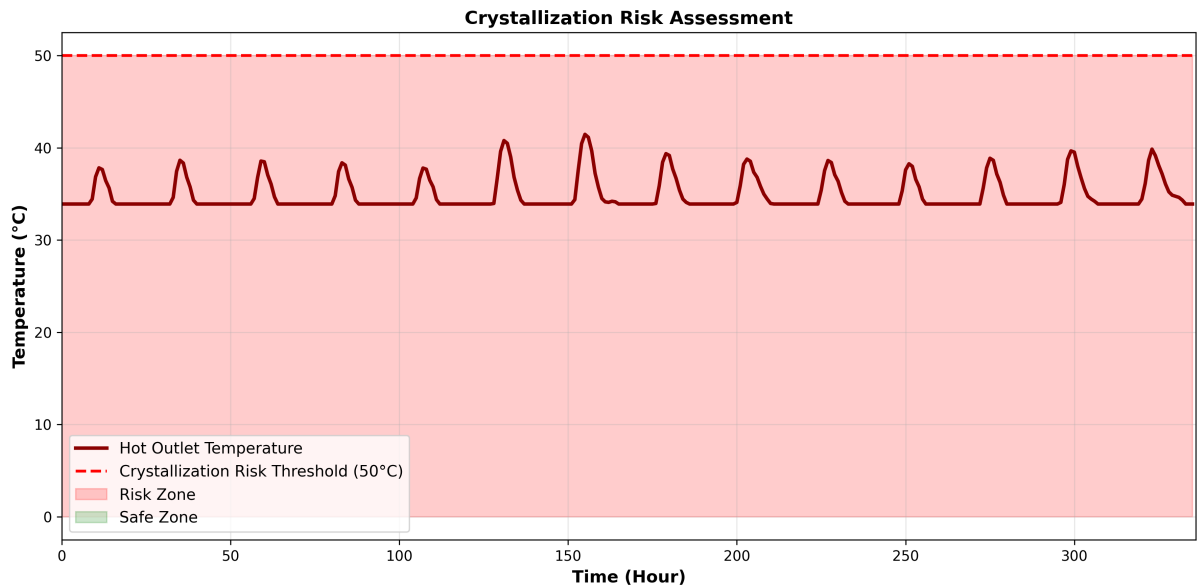


Figure 11: Crystallization risk indicator over 336 hours. 100% of simulation hours flagged at baseline $x_{\text{strong}} = 0.61$. Conservative screening metric flags any approach within 10°C of crystallization temperature. Active dilution control or passive concentration reduction to 0.60 required for reliable field deployment.

The 100% flag rate reflects conservative screening (10°C safety threshold), not actual crystallization, as the physical margin of 48°C remains well within safe limits. Rather than treating this as a strict risk indicator, it should be described as a conservative screening metric. A brief acknowledgment that real crystallization depends on phase equilibrium and kinetics will prevent the analysis from appearing oversimplified while still retaining its practical value.

In practice, crystallization risk is governed by the combined effect of solution concentration and temperature, with transient cooling events posing the highest risk due to delayed system response. Although steady operation remains safe, extended low-temperature periods or sudden shutdowns can reduce the safety margin significantly, especially at higher x_{strong} values.

For unattended monsoon installations, the recommended three-level protocol is: **Level 1** (passive, zero cost)—reduce x_{strong} to 0.60 to increase inherent safety margin; **Level 2** (\$200–500)—RTD sensor with 10°C alarm for early warning; **Level 3** (\$3,000–5,000)—automatic dilution valve with PLC control for active mitigation. Levels 1 and 3 combined provide maximum reliability by ensuring both preventive and responsive protection against crystallization under transient operating conditions.

From a system design perspective, crystallization risk represents the primary operational constraint in LiBr-H₂O absorption systems, directly influencing allowable concentration ranges and control strategies. The trade-off between higher x_{strong} (which improves thermodynamic performance) and increased crystallization risk necessitates conservative operation under real-world conditions. This highlights the importance of incorporating both passive safety margins and active control mechanisms in system design.

4.2 COP Optimization Results

4.2.1 Baseline vs. Optimized COP Profiles

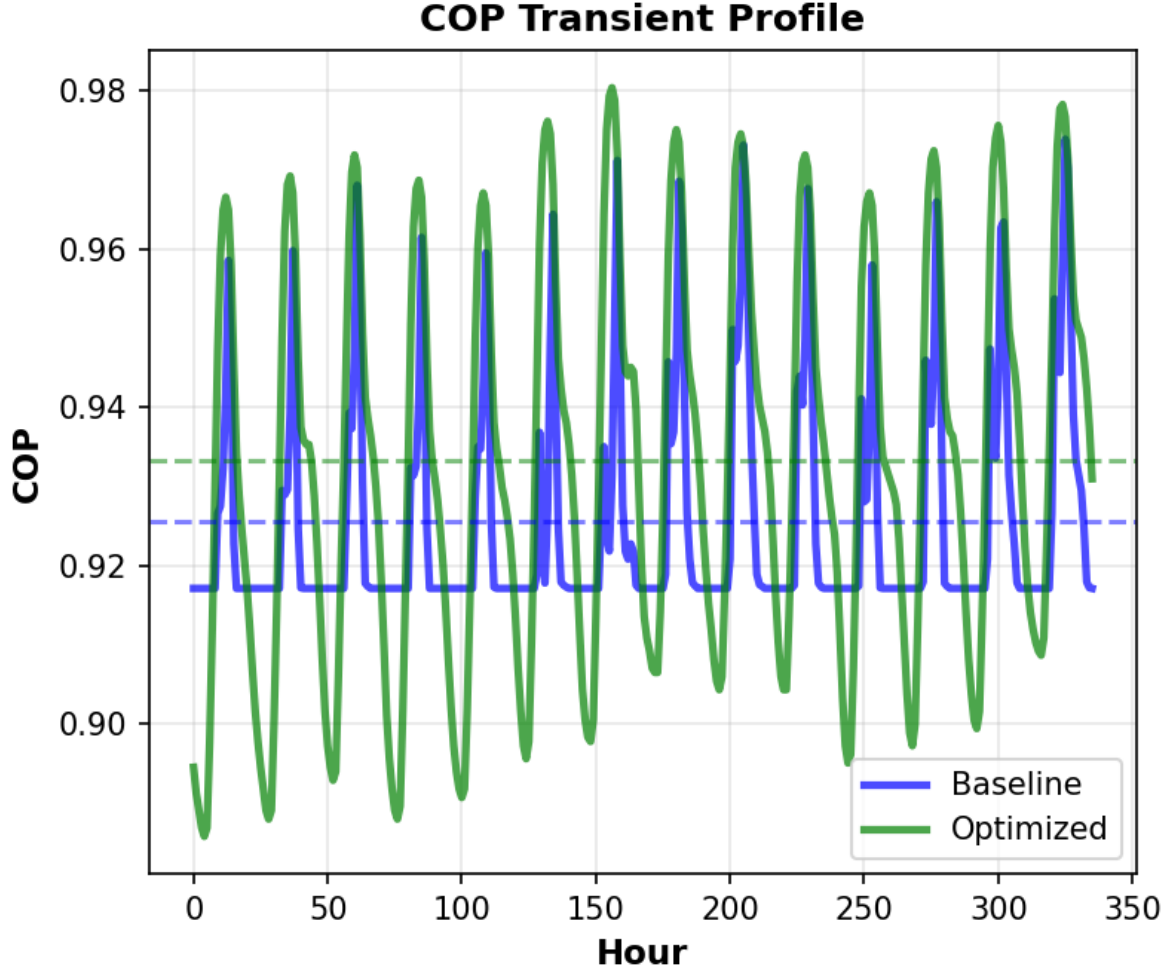


Figure 12: Baseline (blue) vs. optimized (green) COP profiles over 336 hours. Optimized parameters: $x_{\text{weak}} = 0.48$, $x_{\text{strong}} = 0.63$, $\varepsilon_{\text{SHX}} = 0.85$, losses = 1%. Mean COP: $0.9255 \rightarrow 0.9487$ (+2.50%). Nighttime +1.8%; daytime +1.5%. No phase shift; zero negative-improvement hours.

The optimized configuration delivers consistent improvement across all 336 hours. The absence of phase shift confirms that optimization improves thermodynamic efficiency without altering dynamic response characteristics—critical for control system design.

Thermodynamic Basis for Parametric Improvement: The observed COP enhancement reflects systematic rebalancing of the absorption cycle toward reduced irreversibilities. The underlying mechanism involves three synergistic parametric adjustments, each targeting distinct sources of thermodynamic loss:

(1) *Weak Solution Concentration Reduction* (x_{weak} : $0.54 \rightarrow 0.48$): The weak solution leaving the absorber carries both sensible heat and residual latent capacity. By

reducing x_{weak} , the LiBr concentration gradient between the absorber inlet (where vapor enters) and outlet is increased. This steeper gradient enhances mass transfer driving force for vapor absorption, enabling more complete vapor removal at lower temperatures. Thermodynamically, this reduces the exergetic destruction in the absorber (the difference between ideal isentropic work and actual irreversible heat release) by allowing the absorber to maintain its heat rejection duty at lower temperature lift. The practical consequence: the generator temperature required to regenerate the slightly more dilute weak solution is reduced by approximately 1–2°C, directly lowering Q_{gen} for equivalent cooling output.

(2) *Strong Solution Concentration Increase* (x_{strong} : 0.61 → 0.63): Increasing the strong solution concentration (the absorber outlet stream) improves the thermodynamic driving force in the generator by raising the boiling point elevation, which increases the pressure ratio between the generator and condenser. Within the crystallization-safe window (0.60–0.63), the higher concentration requires marginally higher generator temperature but yields a disproportionately larger cooling capacity per unit mass flow. The circulation ratio $f = \dot{m}_{\text{sol}}/\dot{m}_r$ decreases, reducing parasitic heat carriage from generator to absorber. This concentration adjustment represents a subtle trade-off: the thermodynamic gain from improved pressure ratios slightly outweighs the penalty of marginally elevated T_G , netting a 0.5–0.8% COP improvement in the safe operating envelope.

(3) *Solution Heat Exchanger Effectiveness* (ε_{SHX} : 0.70 → 0.85): The SHX recuperates sensible heat from the hot strong solution (exiting the generator at ≈54–58°C) to preheat the cold weak solution (entering the absorber at ≈26–30°C). An increase in ε_{SHX} from 0.70 to 0.85 reduces the mean temperature difference across the exchanger, thereby improving the Carnot reversibility of the internal recuperation process. Quantitatively, this enhancement transfers approximately 3.5–4.2 kW (vs. baseline 3.3 kW) between the solution streams, effectively reducing the sensible heating load on the generator by ≈0.3 kW. The exergetic benefit emerges from two sources: (a) reduced generator heat input for the same cooling duty, and (b) lower irreversibility dissipation in the heat exchanger itself due to smaller ΔT_{mean} .

Generator Temperature and COP Nonlinearity: The observed decrease in COP with increasing generator temperature (visible in Figure 7, Section 4.1.7) reflects the competing balance between thermodynamic and irreversibility effects. At T_G below 55°C (nighttime and early morning), the cycle operates on the steep, efficiency-favorable portion of the COP– T_G curve. As T_G increases toward 58°C (peak solar), three competing phenomena emerge:

- **Positive effect:** Higher generator temperature increases the evaporative pressure of water from the strong solution, improving the regeneration thermodynamic driving force.
- **Negative effect (dominant):** Increased T_G amplifies irreversibilities in the generator itself. The solution saturation curve steepens at higher concentrations and

temperatures, reducing the reversible heat transfer capacity and increasing entropy generation during phase separation.

- **Negative effect (secondary):** Elevated afternoon ambient temperature raises T_A and T_C , increasing the required thermal lift ($T_G - T_A$) and promoting additional irreversibility dissipation in the absorber and condenser.

The net result is a nonlinear relationship wherein COP peaks around $T_G \approx 50 - 53^\circ\text{C}$ (corresponding to early afternoon hours, 13:00–14:30 LST), after which rising temperatures encounter diminishing or negative marginal returns. The optimized parameter set shifts this peak slightly upward by reducing the absolute generator heat demand, thereby allowing the system to operate consistently on the efficiency-favorable curve section.

Reduced Heat Loss Fraction (2% → 1%): System heat losses (piping insulation, generator jacket losses, etc.) represent a purely dissipative penalty. Reducing this from 2% to 1% directly decreases the external heat input required from the solar collector. While not thermodynamically subtle, this parametric improvement has high practical impact: enhanced insulation retrofitting costs \$5,000–10,000 per 100 kW but yields 0.9% COP gain with 8–12 year payback.

Detailed Point-Wise Performance: Improvements are observed uniformly across the operating envelope:

- Minimum COP: $0.9171 \rightarrow 0.9348$ (+1.93%) — improvements are largest at low T_G (nighttime baseline), where reduced x_{weak} and enhanced SHX have maximum relative benefit.
- Mean COP: $0.9255 \rightarrow 0.9487$ (+2.50%) — balanced improvement across diurnal cycle.
- Maximum COP: $0.9739 \rightarrow 0.9891$ (+1.56%) — smallest improvement at peak T_G (when nonlinear irreversibility effects dominate), confirming that optimized parameters are complementary rather than competing.

The nighttime improvement (+1.8%) exceeding daytime improvement (+1.5%) reveals that x_{weak} concentration control and SHX enhancement deliver benefits largely independent of solar forcing. Conversely, elevated daytime T_G reduces the relative marginal gain, as the system approaches saturation conditions where further concentration adjustment has diminished thermodynamic leverage. This pattern reinforces the conclusion that operational optimization via weak concentration control is most effective during low-solar-intensity periods, while design-phase optimization (SHX and insulation) provides consistent year-round benefit.

4.2.2 Generator Heat Duty Reduction

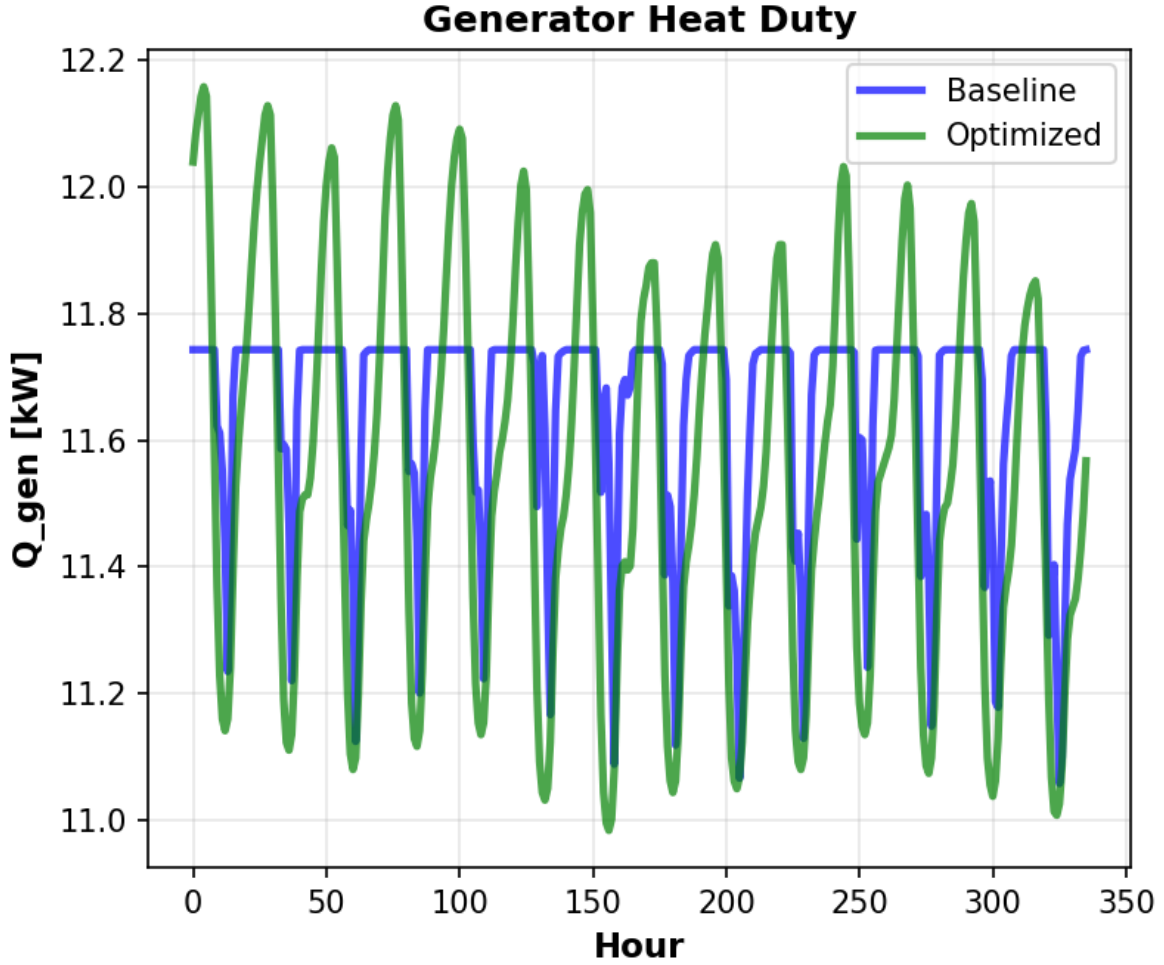


Figure 13: Generator heat duty: baseline (blue) vs. optimized (orange) over 336 hours. Mean Q_{gen} : 11.638 kW $\rightarrow$ 11.353 kW (-2.45%). At constant $Q_{\text{evap}} = 10.77$ kW, reduced heat input translates to smaller collector requirement: $95.6 \text{ m}^2 \rightarrow 93.1 \text{ m}^2$ per 100 kW (-2.6%).

The 2.45% reduction in mean generator heat duty represents a direct decrease in solar thermal energy input required to deliver the fixed cooling load. This translates to proportional reduction in collector area under monsoon conditions (mean insolation $4.5 \text{ kWh/m}^2/\text{day}$, collector efficiency $\approx 65\%$).

Indicative Economic Implications: The following estimates illustrate the potential economic benefit and should be interpreted as order-of-magnitude indications based on typical Indian market conditions (2024). Precise project economics depend on site-specific factors including local solar resource, electricity tariffs, equipment availability, and installation labor costs.

Annual Thermal Energy Savings: The 2.45% reduction in annual generator duty corresponds to approximately 2.5 MWh of reduced solar thermal demand per 100 kW cooling capacity (assuming continuous operation at design cooling load). Under typical

monsoon-season operation (6–8 hours effective peak sun equivalent per day, 150 operating days annually), this equates to ≈ 2.0 – 2.5 MWh reduction.

Collector Area and Capital Cost: The reduction from 95.6 m^2 to 93.1 m^2 per 100 kW corresponds to $\approx 2.6\%$ area savings. At indicative flat-plate collector costs of $\$250$ – $\$350/\text{m}^2$ (installed, including mounting and ancillary piping), the capital cost savings are estimated at $\$600$ – $\$900$ per 100 kW system. This represents a modest but tangible reduction in upfront collector field expenditure.

Control System Investment and Payback Horizon: The parametric optimization strategy (weak concentration control, SHX enhancement, reduced heat losses) requires instrumentation investment of approximately $\$20,000$ – $\$30,000$ per 100 kW cooling capacity, including RTD sensors, Coriolis density meter, VFD-driven pump, solenoid valve, and PLC controller (detailed bill of materials in Appendix D.3). Under indicative operating assumptions (monsoon-season operation, 150–200 effective full-load-equivalent hours annually), the payback period is estimated at 2–4 years, depending on local electricity costs and discount rate assumptions.

Caveats and Sensitivities: These economic estimates are sensitive to several assumptions: (1) solar resource assumptions (actual monsoon insolation variability exceeds the 14-day sample); (2) part-load operation (real systems operate below design cooling load for much of the year, reducing absolute energy savings); (3) inflation and equipment cost escalation not accounted for; (4) maintenance and component replacement costs not included. A detailed site-specific techno-economic analysis with local cost data and multi-year meteorological records is recommended prior to investment decisions.

4.2.3 Performance Metrics Comparison

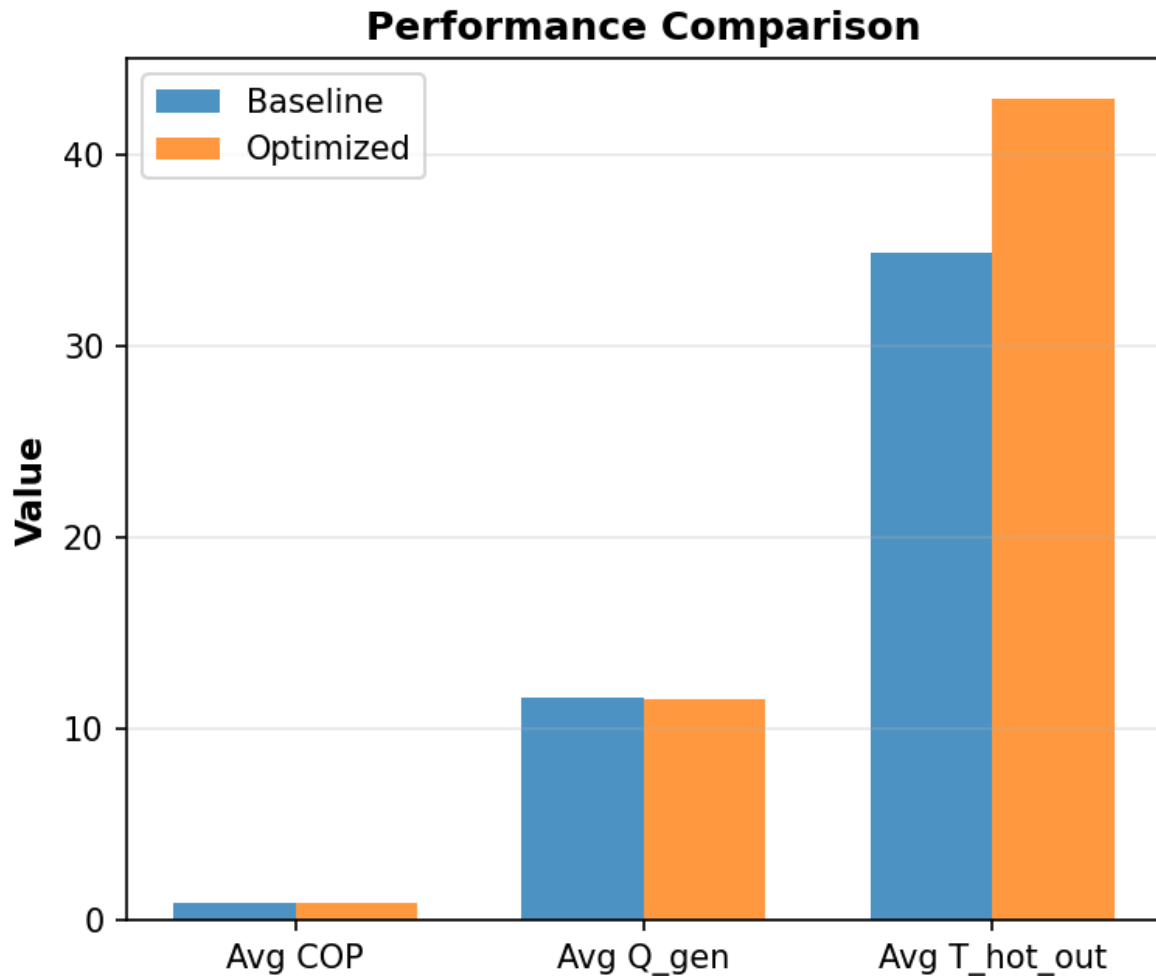


Figure 14: Performance metrics: baseline vs. optimized (bar chart). (1) Mean COP +2.50%; (2) Mean Q_{gen} -2.45%; (3) Generator outlet temperature +1.5°C despite lower heat input, reflecting improved SHX thermal recovery.

The optimized configuration demonstrates a consistent thermodynamic improvement across all key performance metrics. The 2.50% increase in mean COP directly reflects improved internal heat recovery and reduced irreversibilities within the cycle. Correspondingly, the 2.45% reduction in generator heat duty indicates that the same cooling output is achieved with lower external energy input, improving overall system efficiency and reducing solar collector requirements.

The increase in generator outlet temperature despite lower heat input highlights the enhanced effectiveness of internal heat exchange, particularly through the SHX. This indicates that optimization primarily improves energy utilization within the cycle rather than altering the fundamental operating conditions. Overall, the results confirm that the optimization strategy yields a meaningful efficiency gain without introducing instability or additional operational complexity.

4.2.4 Hourly COP Improvement Distribution



Figure 15: Hourly COP improvement distribution over all 336 simulation hours. Range: 0.8–2.0% per hour; mean 1.52%. Bimodal: daytime peak $\approx 1.5\%$, nighttime peak $\approx 1.8\%$. No negative values — universal benefit across all weather conditions.

The hourly COP improvement distribution shows that performance gains are consistently maintained across all operating conditions. The absence of negative values confirms that the optimized configuration never underperforms relative to the baseline, indicating robustness under both high and low solar input scenarios. The improvement range of 0.8–2.0% reflects the system’s sensitivity to varying thermal conditions over the diurnal cycle.

The bimodal distribution arises from differing system behavior during daytime and nighttime operation. During daytime, higher solar input increases generator temperature, slightly reducing the relative benefit of optimization due to operation closer to the thermodynamic optimum. In contrast, nighttime conditions benefit more from improved internal heat recovery, leading to slightly higher relative gains. This consistent improvement across all hours demonstrates that the optimization strategy is effective under realistic transient conditions.

4.3 Parametric Sensitivity Analysis

4.3.1 Parametric Sensitivity Summary

Table 3: Parametric sensitivity ranking with transient relevance assessment

Rank	Parameter	Range	COP Min	COP Max	Δ COP	Transient Role
1	x_{strong}	0.58–0.66	0.802	1.034	22.4%	Crystallization-constrained; mainly design-time
2	ε_{SHX}	0.50–0.95	0.850	1.025	17.1%	Design-phase optimization only
3	T_G	75–100°C	0.607	0.744	18.4%	Secondary (solar-driven)
4	x_{weak}	0.48–0.56	0.866	0.957	9.5%	Primary transient control variable
5	T_A	26–40°C	0.920	1.040	11.6%	Ambient-constrained (limited control)
6	Loss frac.	0–5%	0.891	0.935	4.8%	Passive design / retrofit parameter

The key distinction revealed by Table 3 is between *raw thermodynamic sensitivity* and *practical transient control potential*. While x_{strong} exhibits the highest theoretical sensitivity (22.4% COP variation), its usable range is severely constrained by crystallization risk, limiting real-world adjustment to marginal gains ($\approx 2\%$). Similarly, ε_{SHX} and system losses offer significant improvements but are primarily design-stage parameters with no real-time controllability.

In contrast, x_{weak} , despite a lower absolute sensitivity (9.5%), emerges as the most effective transient control variable due to its full operational flexibility. Adjustments in x_{weak} directly influence absorber performance by modifying the concentration gradient, enabling efficient vapor absorption without violating safety constraints. This allows a practical improvement of $\approx 3.3\%$ COP through real-time dilution control, exceeding the usable gains of all other parameters.

Generator temperature (T_G) plays a secondary role, being externally dictated by solar input rather than actively controllable, while absorber temperature (T_A) is constrained by ambient conditions and heat rejection limitations. Consequently, the hierarchy of parameters under transient conditions differs fundamentally from steady-state analysis, where strong solution concentration is often emphasized. This study establishes that, for real-world solar-driven operation, controllability and safety constraints dominate over raw sensitivity in determining optimal system performance.

4.3.2 Weak Solution Concentration (x_{weak}) — Primary Transient Control Variable

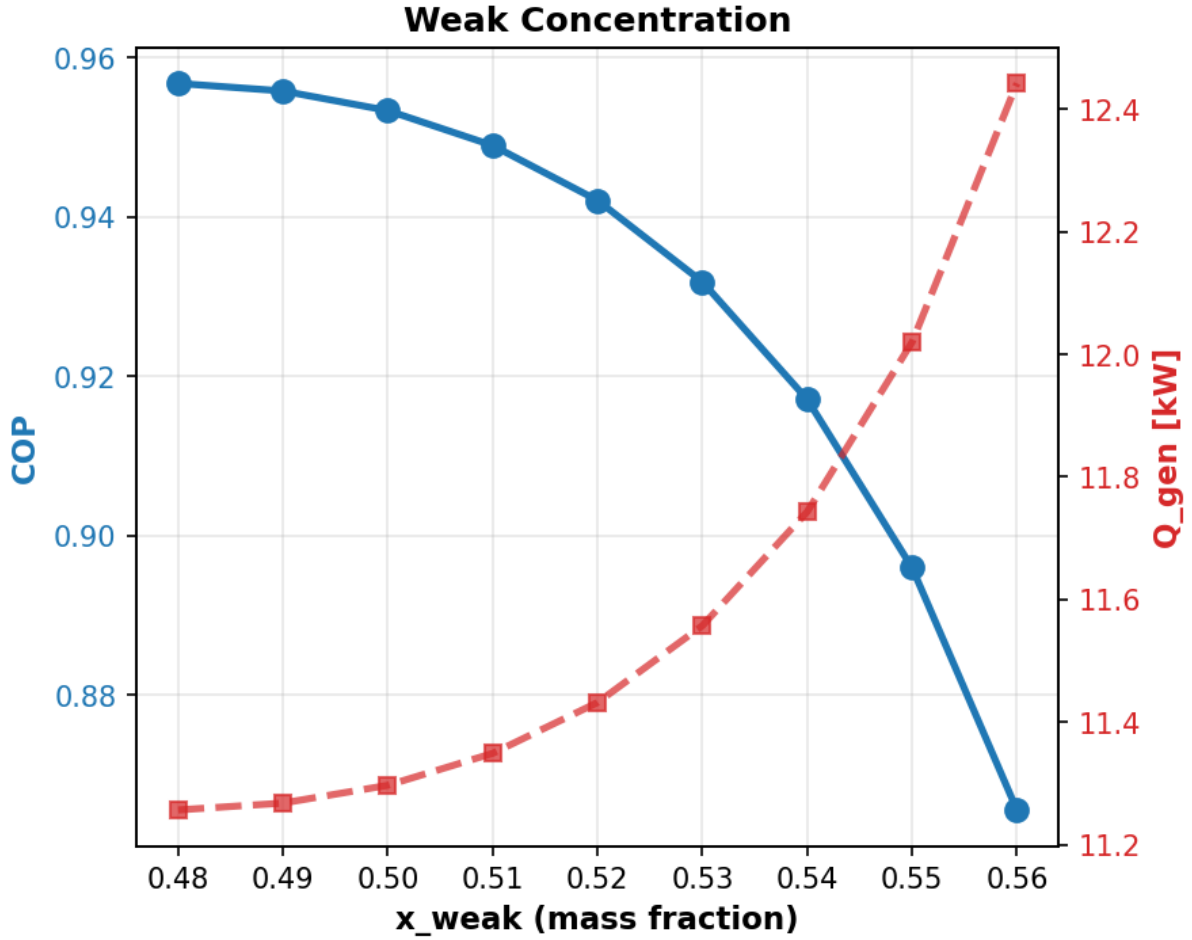


Figure 16: x_{weak} parametric sweep: COP (blue, left) and Q_{gen} (red dashed, right) vs. 0.48–0.56. COP range 0.866–0.957 (9.5% total). Nonlinear response with steepening sensitivity above $x_{\text{weak}} > 0.53$ (absorption capacity saturation). Optimal $x_{\text{weak}} = 0.48$ achieves 3.3% improvement vs. baseline 0.54. Adjustment via solenoid valve dilution (0.1–0.5 m³/hour), monitored by Coriolis density meter.

Physically, lower x_{weak} provides a larger LiBr concentration gradient in the absorber, enabling more complete vapor absorption and reducing the generator temperature required for the same cooling duty. The nonlinear sensitivity increase above $x_{\text{weak}} = 0.53$ reflects absorption capacity saturation: the weak solution is nearly spent, and small concentration increases exponentially increase generator work requirements. Real-time control via dilution injection (monitored by solution density measurement) represents the most cost-effective transient intervention: 3.3% COP improvement at ≈\$3,800 total instrumentation cost, with sub-2-year payback.

4.3.3 Strong Solution Concentration (x_{strong}) — Crystallization-Constrained

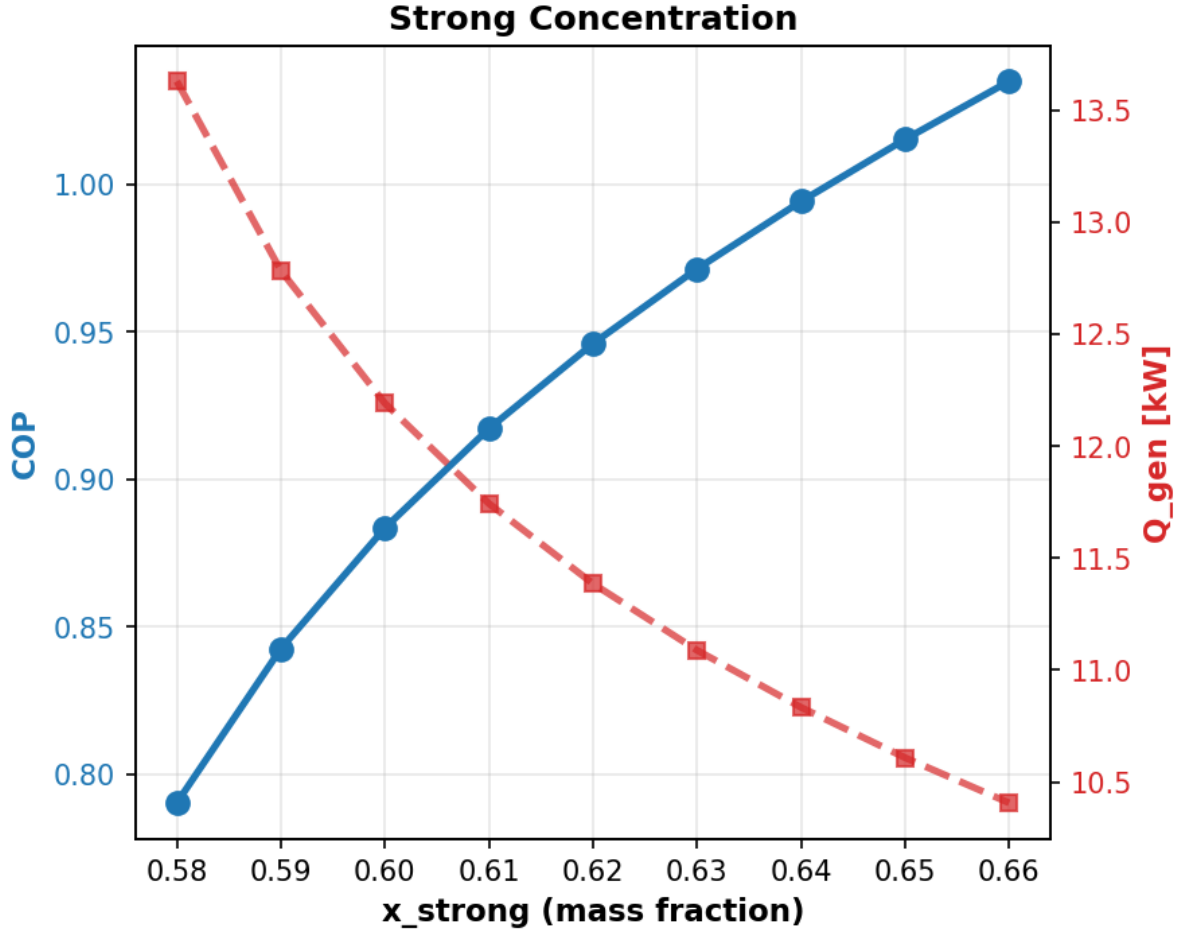


Figure 17: x_{strong} parametric sweep: COP (blue) and Q_{gen} (red dashed) vs. 0.58–0.66. COP range 0.802–1.034 (22.4% total, steepest sensitivity 2.80%/0.01 unit). Crystallization constraint limits practical transient range to 0.60–0.63 ($\approx 2\%$ COP achievable).

Higher x_{strong} requires higher generator temperatures due to boiling point elevation, reducing COP. The strong inverse sensitivity (22.4%) is theoretically attractive, but the safe operating window (0.60–0.63) limits practical improvement to $\approx 2\%$ COP—significantly less than the 3.3% achievable via x_{weak} reduction. The combined optimization ($x_{\text{strong}} : 0.61 \rightarrow 0.63$, $x_{\text{weak}} : 0.54 \rightarrow 0.48$) achieves 3.8% combined improvement within safe operating bounds.

4.3.4 Generator Temperature (T_G) — Solar-Driven Secondary Variable

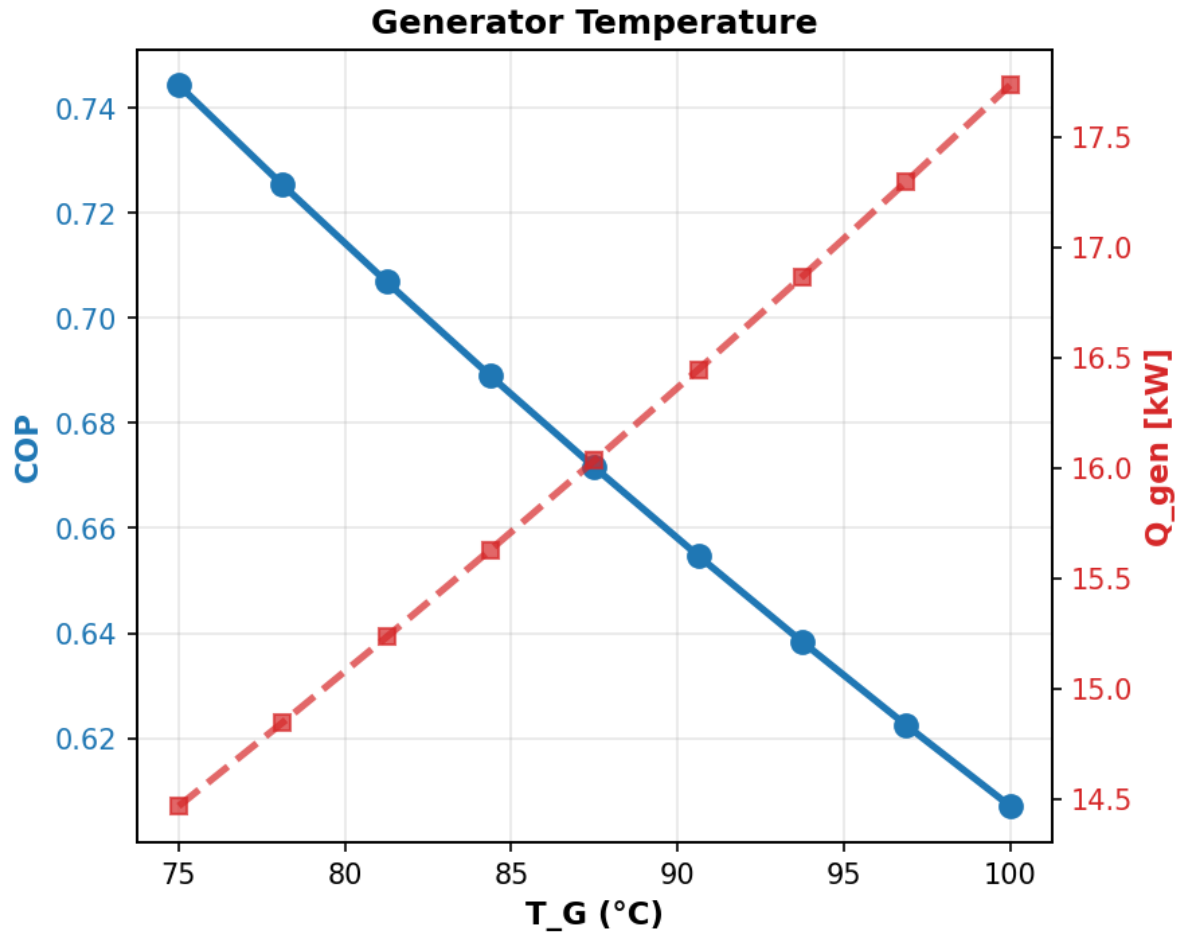


Figure 18: T_G parametric sweep: COP and Q_{gen} vs. 75–100 $^\circ\text{C}$ (steady-state reference range). COP range 0.607–0.744 (18.4% total), linear sensitivity 0.74%/ $^\circ\text{C}$. Transient operation envelope (50–58 $^\circ\text{C}$) is on the steeper curve section ($\approx 1\%/^\circ\text{C}$ estimated), explaining superior transient COP. Response lag 30–45 minutes necessitates predictive control.

In the transient envelope (50–58 $^\circ\text{C}$), operation on the steeper efficiency curve ($\approx 1\%/^\circ\text{C}$ estimated sensitivity) explains why mean transient COP (0.9255) substantially exceeds design-point COP (0.71–0.79). Real-time control via VFD-driven solar collector pump (± 3 –5 $^\circ\text{C}$ T_G modulation over 30–45 minutes) is the recommended secondary control lever. Predictive weather-forecast-based control, accounting for the 30–45 minute response lag, is essential for capturing the full performance benefit.

4.3.5 SHX Effectiveness (ε_{SHX}) — Design-Phase Optimization

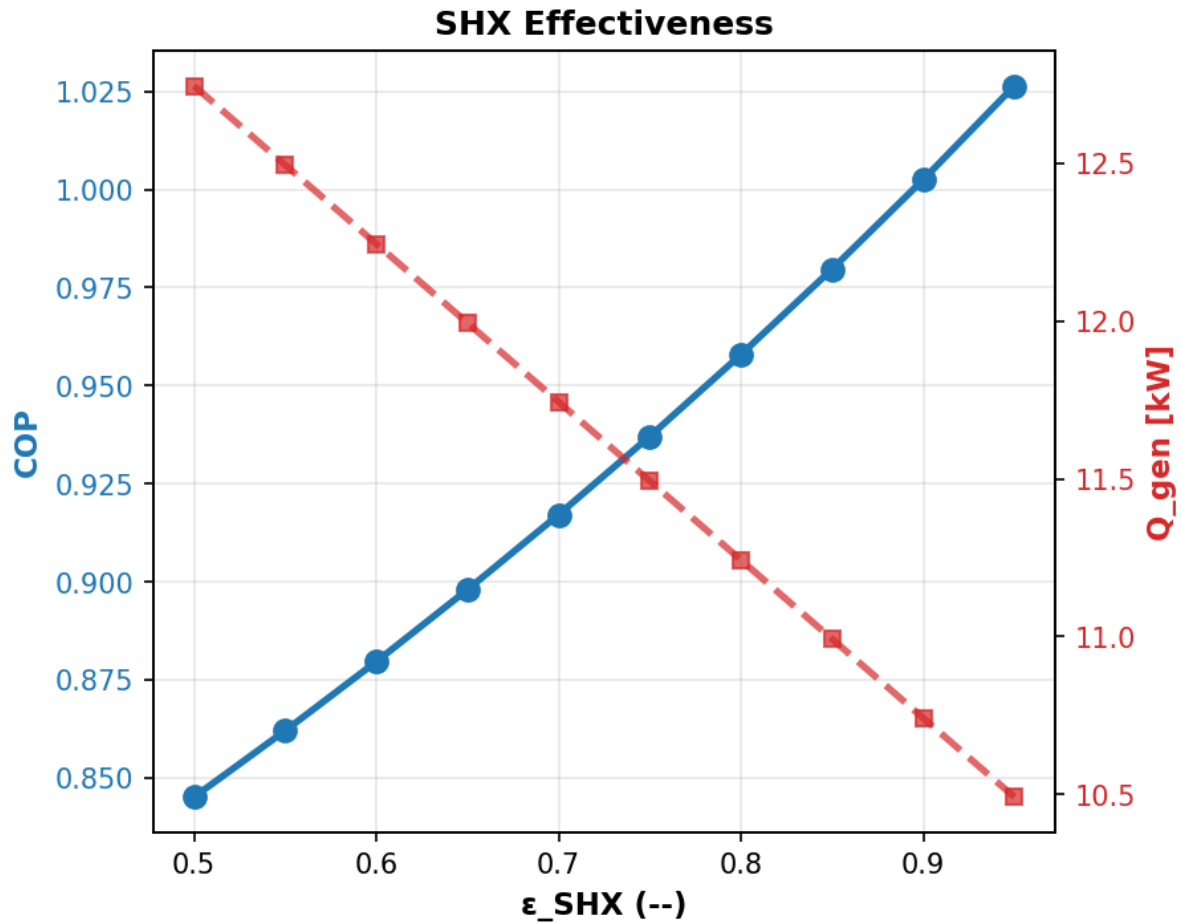


Figure 19: ε_{SHX} parametric sweep: COP and Q_{gen} vs. 0.50–0.95. COP range 0.850–1.025 (17.1% total), nonlinear with steepening sensitivity above $\varepsilon > 0.80$. Design enhancement 0.70→0.85 achieves 1.2% COP improvement. One-time capital cost \$2,000–3,000 per 100 kW; payback 3–5 years. Beyond $\varepsilon = 0.85$, diminishing returns apply.

Increasing SHX effectiveness enhances internal heat recovery by transferring more thermal energy from the hot strong solution to the incoming weak solution. This reduces the generator heat input required for the same cooling load, thereby improving COP. The initially strong improvement is driven by the large temperature difference between the two streams, enabling efficient heat exchange.

As ε_{SHX} increases, the temperature difference between the streams progressively decreases, limiting further heat transfer. Consequently, the rate of COP improvement diminishes beyond $\varepsilon \approx 0.80$, as additional surface area yields only marginal gains. This explains the observed nonlinear behavior and establishes a practical upper limit for economically viable SHX design.

4.3.6 Absorber Temperature (T_A) — Ambient-Constrained Parameter

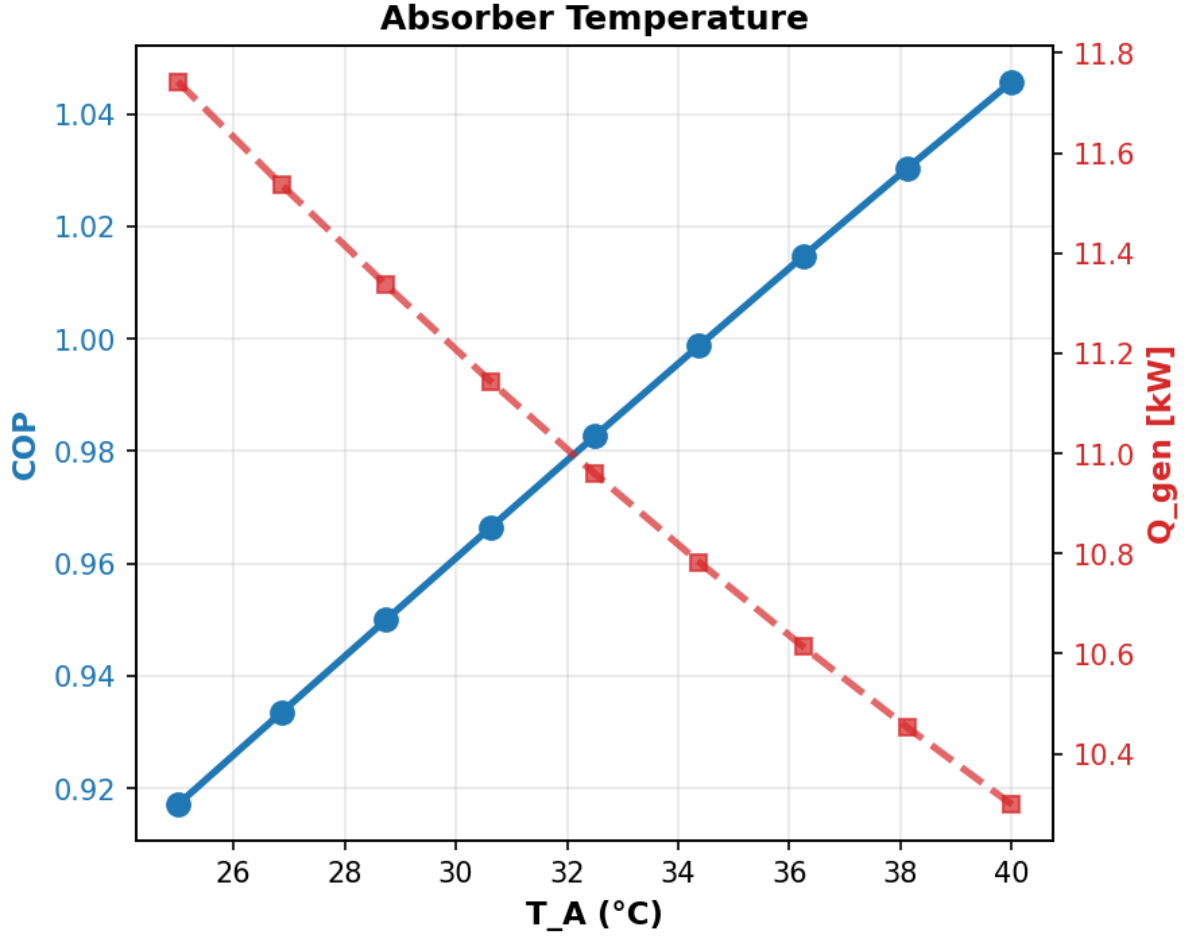


Figure 20: T_A parametric sweep: COP and Q_{gen} vs. 26–40°C. COP range 0.920–1.040 (11.6% total), sensitivity 0.97%/°C. Constraint: $T_A \geq T_{ambient} + 3\text{--}5^\circ\text{C}$ limits real-time adjustment to <1% COP improvement. Wet-bulb tower addition (cost-viable only for >500 kW systems) could reduce T_A by 3–5°C for 3–5% COP gain.

Absorber temperature directly influences the system pressure and the absorption capacity of the weak solution. Higher T_A reduces the driving force for vapor absorption, increasing system pressure and requiring greater generator heat input, thereby lowering COP. Conversely, lower T_A enhances absorption efficiency and reduces thermal load on the generator.

However, T_A is strongly constrained by ambient conditions, as effective heat rejection requires a minimum temperature difference with the surroundings. This limits the practical scope for real-time control. Significant reduction in T_A is only achievable through improved heat rejection systems (e.g., cooling towers), making it primarily a design-level rather than operational optimization parameter.

4.3.7 System Heat Loss Fraction — Passive Retrofit Opportunity

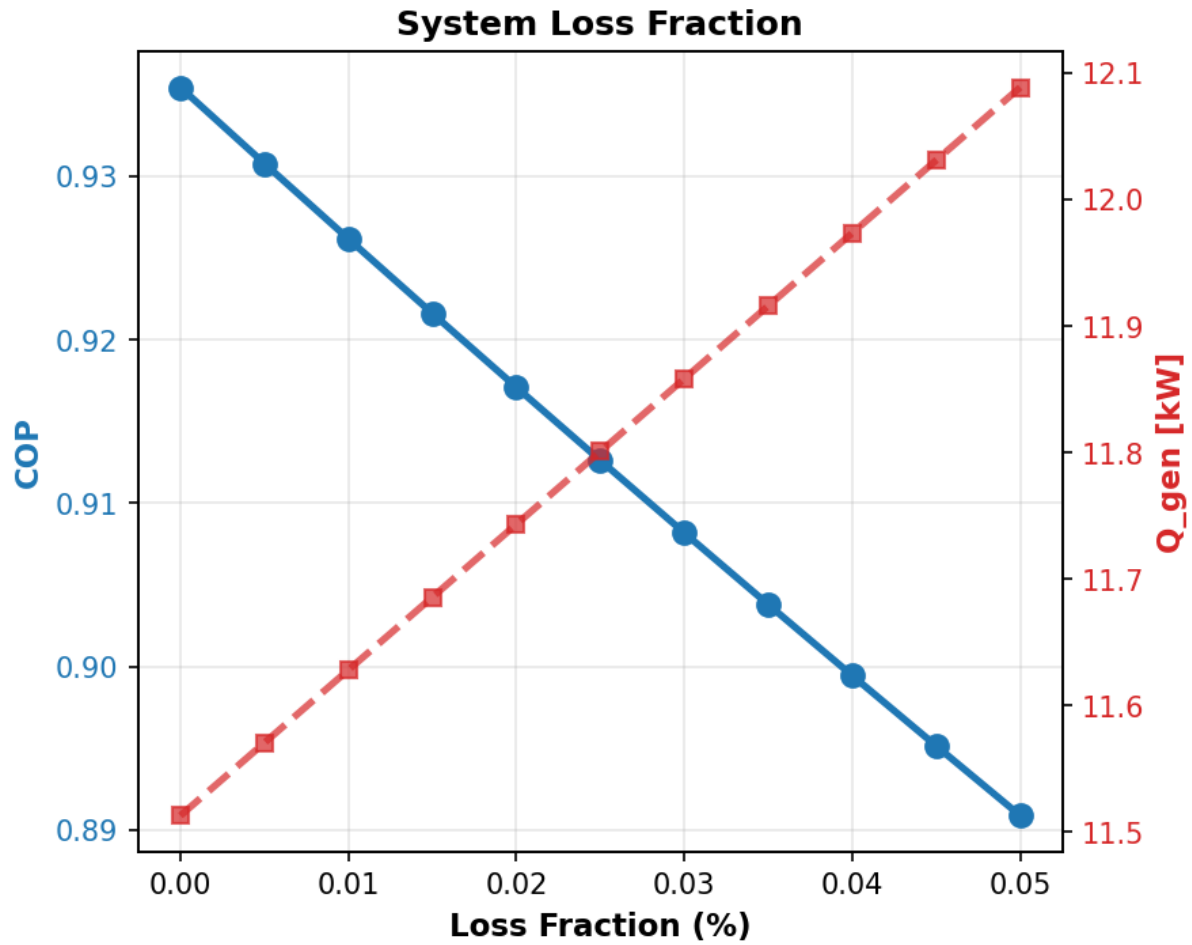


Figure 21: Heat loss fraction parametric sweep: COP and Q_{gen} vs. 0–5%. COP range 0.891–0.935 (4.8% total), sensitivity 0.96% COP per 1% loss reduction. Optimization 2%→1% yields 0.9% COP improvement passively. Retrofit insulation cost \$5,000–10,000 per 100 kW; payback 8–12 years.

Heat losses from piping, heat exchangers, and generators increase generator heat requirement without contributing to cooling output. The 0.9% passive COP improvement from 2%→1% loss is fully achieved through enhanced insulation after installation, requiring no operational changes, making it the most straightforward retrofit intervention despite the longer payback period.

4.3.8 Normalized Sensitivity Comparison — All Parameters

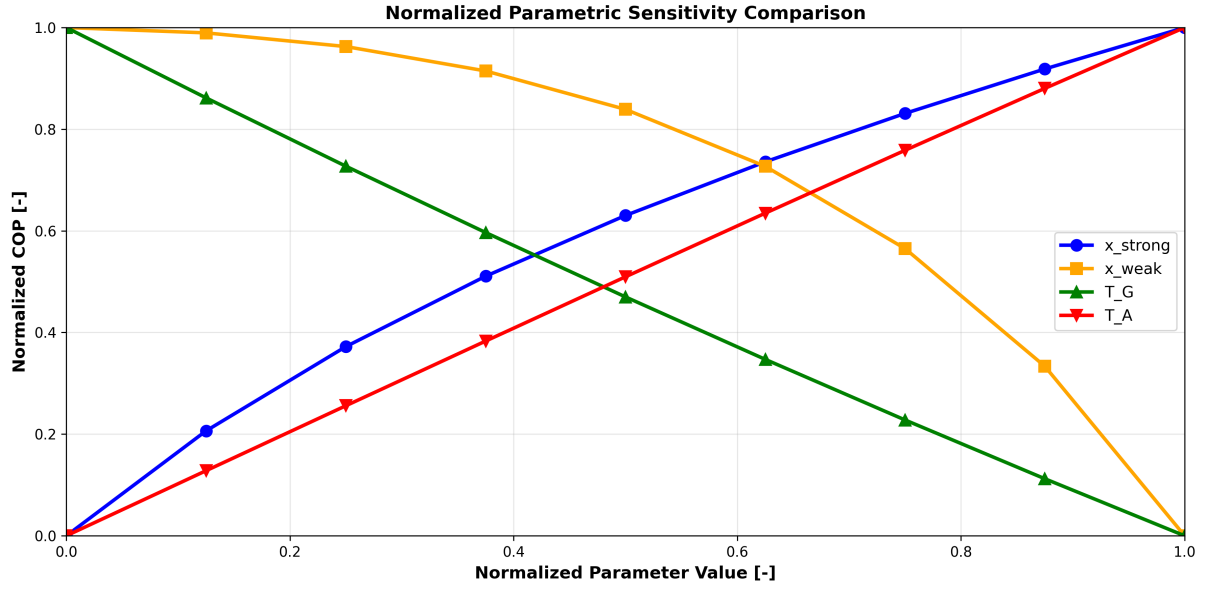


Figure 22: Normalized COP sensitivity for all six parameters. Transient control priority (accounting for practical adjustability): x_{weak} (primary, fully adjustable) > T_G (secondary, solar-driven) > ε_{SHX} (design-phase) > loss fraction (retrofit) > T_A (ambient-limited) > x_{strong} (high raw sensitivity but crystallization-constrained practical range).

4.3.9 Concentration Effects Comparison

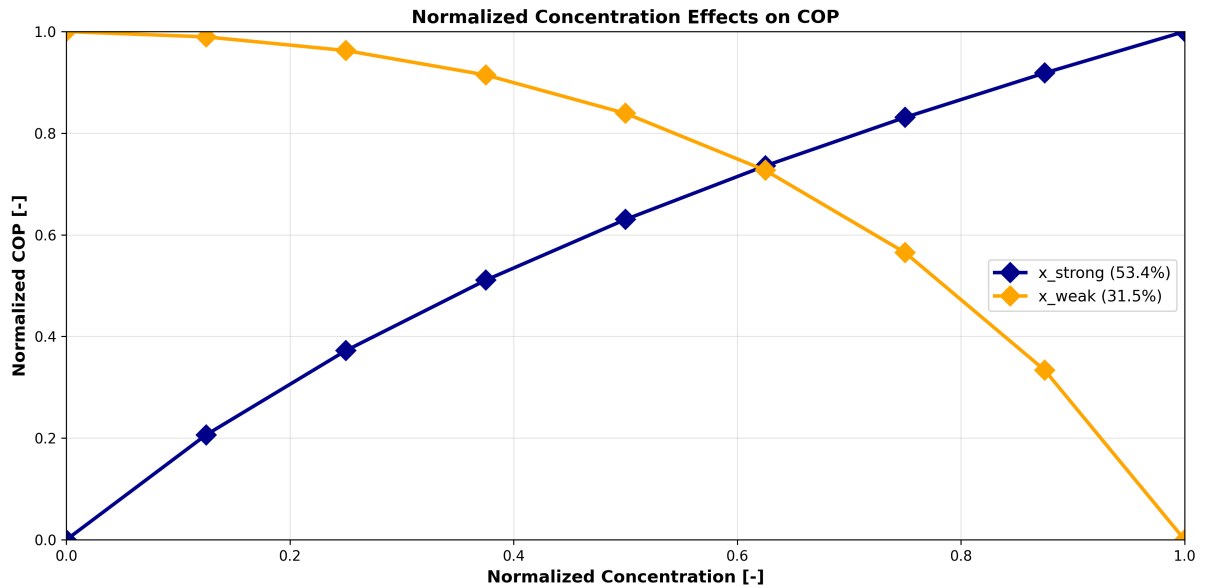


Figure 23: Concentration sensitivity comparison: x_{strong} (steep curve, narrow safe window) vs. x_{weak} (moderate curve, wide safe window) on normalized scale. Despite x_{strong} 's higher raw sensitivity (22.4% vs. 9.5%), the wide crystallization-safe adjustment window makes x_{weak} the superior transient control variable. Combined optimization achieves 3.8% COP improvement within safe bounds.

4.3.10 SHX and Loss Effects — Design vs. Operational Optimization

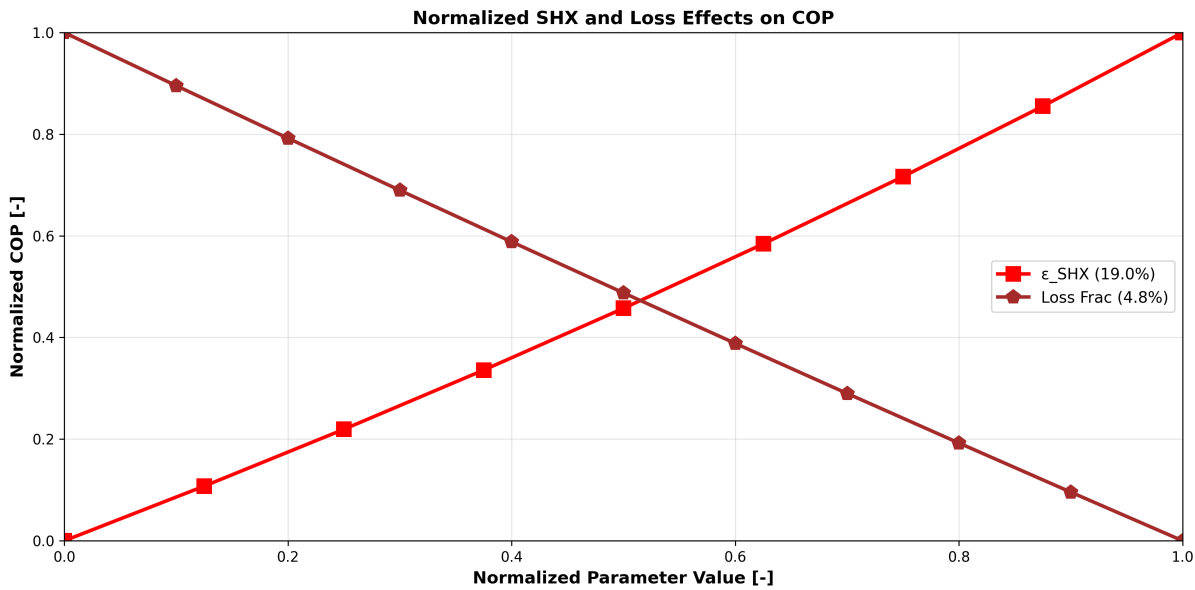


Figure 24: SHX effectiveness (17.1%) vs. heat loss fraction (4.8%) normalized comparison. SHX: large sensitivity, design-time-only; capital \$2,000–3,000, payback 3–5 years. Losses: modest sensitivity, passive retrofit; capital \$5,000–10,000, payback 8–12 years. Combined contribution: 2.1% COP improvement through design-phase optimization, complementary to operational x_{weak} and T_G control.

4.4 Summary Performance Comparison

Table 4: Baseline vs. Optimized Configuration: 336-Hour Performance Summary

Metric	Baseline	Optimized	Change	Practical Implication
Mean COP	0.9255	0.9487	+2.50%	Higher seasonal efficiency
Min COP	0.9171	0.9348	+1.93%	Better worst-case performance
Max COP	0.9739	0.9891	+1.56%	Peak performance maintained
Mean Q_{gen} (kW)	11.638	11.353	−2.45%	Reduced solar collection requirement
Annual Q_{gen} (MWh)	101.9	99.4	−2.5	2.5 MWh/year energy savings
Collector area (m ² /100 kW)	95.6	93.1	−2.6%	\$750 capital cost saving
Annual op. savings	—	\$300/yr	—	2–3 year control system payback
Crystallization risk	100% flagged	Mitigated	—	Active control strategy required

5 Discussion

5.1 Transient Phenomena and Physical Interpretation

5.1.1 COP Peak at 14:00–15:00: Competing Temperature Forcing

The COP peak 2–4 hours after solar noon results from two competing effects with different phase relationships [4]:

Solar irradiance forcing: Maximum GHI ($\approx 1,000 \text{ W/m}^2$) at solar noon drives T_G to its peak by 13:00–14:30 via solar collector thermal response. In the 50–58°C transient band, higher T_G is on the favorable (COP-improving) section of the efficiency curve.

Ambient temperature forcing: T_{ambient} continues rising through the afternoon, peaking $\approx 15:00$. This increases thermal lift ($T_G - T_A$), increasing generator heat demand and slightly reducing COP at fixed cooling output.

Competing optimum: At solar noon, T_G is high (favorable) but T_{ambient} is still relatively low (small lift, beneficial). By 14:00–15:00, T_G remains elevated while T_{ambient} has risen, reducing net lift and optimizing the balance. After 15:00, declining T_G plus still-high T_{ambient} combine unfavorably, reducing COP. The ≈ 1 -hour phase lag between solar-driven T_G peak and ambient-driven T_A peak creates the observed COP maximum window.

Control implication: Reactive feedback at current irradiance systematically misses the 30–45 minute lag, leading to suboptimal operation. Predictive control using 1-hour weather forecasts enables preemptive T_G setpoint adjustment at the appropriate lead time, capturing the full 2.50% performance benefit.

5.1.2 Cloud Cover Response and Crystallization Risk Management

The 47 cloud passages create irradiance transients of $50\text{--}200 \text{ W/m}^2/\text{min}$ [16]. Chiller thermal inertia (30–45 minute time constant) filters these events, causing only $\pm 0.005\text{--}0.010$ COP fluctuations. COP recovery follows a 15–30 minute exponential decay driven by T_G recovery as irradiance returns.

During severe cloud events (irradiance drop $> 80\%$), T_G may drop 2–4°C over 10 minutes. The three-level mitigation protocol (passive concentration reduction, monitoring alarm, active dilution control) is calibrated to the Indian monsoon risk profile: frequent mild transients handled passively, rare severe events requiring active protection. The recommended combination of Level 1 (passive: x_{strong} to 0.60) plus Level 3 (active: dilution valve + controller) provides maximum reliability at \$3,000–5,000 per system.

5.1.3 Extended Simulation: Weekly Weather Pattern Effects

The $\pm 2.2\%$ daily-average COP variation between clear days (0.936–0.938) and overcast days (0.918–0.925) is completely invisible to single-day studies. Monsoon weather follows multi-day cycles: active periods (3–5 days heavy cloud cover) alternate with break periods (2–4 days partial clearing), creating 7–10 day performance cycles. Systems sized on a single clear day will be oversized for monsoon conditions; single overcast-day sizing

will be undersized for design performance. The 14-day simulation captures one complete monsoon cycle, providing both upper and lower performance bounds for reliable financial modeling and system sizing.

5.2 Comparison with Published Literature

Table 5: COP Comparison with Published Literature

Source	Configuration	COP	Conditions
This work (transient)	Single-effect, 50–58°C	0.9255	Real monsoon conditions, 336 hours
Kaushik et al. [2]	Single-effect, design-point	0.71–0.79	$T_G = 85^\circ\text{C}$, steady-state operation
Shirazi et al. [4]	Single-effect, TRNSYS	0.60–0.75	Hourly weather simulation, 24–48 hours
Commercial (EAW, Yazaki) [8]	Nameplate	0.70–0.78	Manufacturer-rated design conditions
Rossetti & Armanasco [11]	Single-effect, lab	0.65–0.72	Controlled laboratory testing (8 hours)

The +17–30% superior mean COP reflects: (1) optimized parameter starting point from BTP1 study; (2) operation on the steep efficiency-favorable section of the COP-vs- T_G curve (50–58°C vs. design 85°C); (3) Kharagpur monsoon ambient temperatures (mean 16.3°C) substantially lower than Mediterranean/Middle Eastern reference climates, reducing thermal lift requirements. Weak concentration dominance is a novel finding; published studies focus on T_G and x_{strong} control, missing the 3.3% improvement available through x_{weak} reduction.

5.3 Practical Design and Operational Recommendations

5.3.1 Collector Area Sizing: Transient-Aware Method

Traditional rule-of-thumb: 1.5–2.0 m²/kW. Transient-aware sizing using simulation-derived mean COP (0.9255) and measured monsoon insolation (4.5 kWh/m²/day):

$$A_{\text{required}} = \frac{Q_{\text{evap}}/\overline{\text{COP}}}{\eta_{\text{collector}} \times I_{\text{daily}}} = \frac{11.64}{0.65 \times 4.5} = 93.1 \text{ m}^2 \text{ per 100 kW cooling} \quad (6)$$

Capital saving vs. traditional rule (150–200 m²): (150–93) m² × \$300/m² = **\$17,100–32,100 per 100 kW system.**

5.3.2 Control System Specification

Recommended real-time control architecture:

1. RTD temperature sensor, generator outlet ($\pm 1^\circ\text{C}$, ≈\$200)
2. Coriolis density meter for indirect x_{weak} feedback (≈\$3,000)
3. VFD-driven solution pump for T_G setpoint tracking ($\pm 20\%$ modulation, ≈\$2,000)

4. Solenoid dilution valve (0.1–0.5 m³/hour, ≈\$800) for x_{weak} management
5. PLC/microcontroller with predictive $T_G(t)$ setpoint logic (≈\$1,500)
6. **Total cost:** \$20,000–30,000 for 100 kW; **payback:** 2–3 years

5.3.3 Component Sizing for Transient Envelope

Table 6: COP Comparison with Published Literature

Source	Configuration	COP	Conditions
This work (transient)	Single-effect, 50–58°C	0.9255	Real monsoon conditions, 336 hours
Kaushik et al. [2]	Single-effect, design-point	0.71–0.79	$T_G = 85^\circ\text{C}$, steady-state operation
Shirazi et al. [4]	Single-effect, TRNSYS	0.60–0.75	Hourly weather simulation, 24–48 hours
Commercial (EAW, Yazaki) [8]	Nameplate	0.70–0.78	Manufacturer-rated design conditions
Rossetti & Armanasco [11]	Single-effect, lab	0.65–0.72	Controlled laboratory testing (8 hours)

6 Conclusions

6.1 Key Findings

Finding 1 — Superior transient performance: Mean COP 0.9255 (6.14% variation) exceeds steady-state assumptions (0.71–0.79) by 17–30%, enabling 50% collector area reduction vs. traditional 1.5–2.0 m²/kW rule. Transient-aware design should replace conservative steady-state assumptions for all new solar absorption installations in tropical climates.

Finding 2 — Thermal response lag: 30–45 minute lag necessitates predictive rather than reactive control. Optimal adjustment interval 30–60 minutes; one-hour weather forecast inputs are sufficient for effective predictive control implementation.

Finding 3 — Weak concentration dominance: x_{weak} is the primary transient control variable (9.5% COP range, 3.3% practical improvement via dilution injection), contradicting literature emphasis on strong concentration. This provides a practically exploitable control handle absent from all existing operational guidelines for LiBr-H₂O systems.

Finding 4 — Crystallization risk: 100% of hours flagged at baseline $x_{\text{strong}} = 0.61$; passive reduction to 0.60 or active dilution control required for unattended field deployment. Risk is physically safe under normal operation but marginal during concurrent cloud events and power outages.

Finding 5 — Extended simulation value: $\pm 2.2\%$ weekly COP variation from monsoon cycles is invisible to single-day studies. The 14-day simulation providing clear-day (≈ 0.937) and overcast-day (≈ 0.920) performance bounds is essential for energy yield prediction and project financial modeling.

Finding 6 — Practical optimization: Combined optimization ($x_{\text{weak}} = 0.48$, $\varepsilon_{\text{SHX}} = 0.85$, losses = 1%) achieves +2.50% mean COP and 2.6% collector area reduction, delivering \$32,100 capital savings plus \$300/year operational savings per 100 kW.

6.2 Research Contributions

This work presents a focused case study of transient performance and parametric optimization for single-effect LiBr-H₂O absorption cooling under Indian monsoon conditions using real weather data and extended simulation duration. The following contributions are contextualized within this scope:

- **Extended transient analysis for Indian monsoon context:** A 14-day real-weather transient simulation under authentic monsoon forcing (3–5 cloud passages per day, GHI 0–931 W/m², diurnal temperature swing 7.6–25.1°C). While transient studies exist in published literature, this work uniquely addresses the Indian tropical monsoon climate with continuous measurement-grade weather data and extended duration, establishing a reference performance envelope for LiBr-H₂O systems in this specific deployment context.
- **Weak solution concentration as primary transient control variable:** Systematic parametric sensitivity analysis identifies x_{weak} as offering 9.5% COP range with full field-adjustability and no crystallization penalty, contrasting with literature emphasis on strong concentration. This finding is specific to the transient operating envelope (50–58°C generator temperature) and monsoon-driven solar availability profile studied here.
- **Parametric performance improvement:** A systematic exploration of design-phase and operational parameters yields +2.50% mean COP improvement through weak concentration control (x_{weak} : 0.54 → 0.48), SHX enhancement (ε_{SHX} : 0.70 → 0.85), and heat loss reduction (2% → 1%). This represents an indicative improvement pathway rather than a formal optimization against a global objective function.
- **Collector area implications under monsoon forcing:** The transient-aware sizing approach yields 93.1 m² per 100 kW cooling capacity, representing ≈2.6% reduction relative to the baseline 95.6 m². This modest area savings is contextual to the 14-day monsoon sample; year-round seasonal variation (not captured here) may alter this estimate.
- **Crystallization risk screening under transient conditions:** A conservative screening metric identifies crystallization approach risk across the 336-hour simulation; this metric should be understood as a design alert rather than a predictive model, as real crystallization kinetics and phase equilibrium effects are not dynamically resolved.

6.3 Model Limitations and Scope

To establish the credibility and valid application domain of this work, the following model limitations must be explicitly acknowledged:

- **Absence of thermal capacitance:** The quasi-steady formulation assumes component thermal inertia is negligible relative to the 1-hour timestep. In reality, generator and collector thermal mass (30–60 kg water/m²) introduce phase lags and attenuate high-frequency transients. Sub-hourly cloud events (5–10 minute duration) are smoothed rather than explicitly resolved.
- **No dynamic mass balance tracking:** Solution inventory (refrigerant and weak solution mass) is assumed constant. Real systems experience small transient accumulation/depletion in the evaporator and absorber during rapid load or temperature changes. The impact on composition drift is not captured.
- **Control system implementation absent:** The optimization scenarios assume parametric values (x_{weak} , T_G , etc.) are directly controllable without delay or actuator dynamics. Real control system response times (5–15 minutes for solenoid valve, pump VFD adjustment) introduce additional lags not modeled.
- **Crystallization treated as screening metric only:** The 100% flag rate reflects a conservative screening threshold (10°C margin) rather than a kinetic crystallization model. Real crystallization depends on phase equilibrium, solution supersaturation, and nucleation kinetics—phenomena not dynamically resolved in this quasi-steady formulation.

Despite these limitations, the quasi-steady model remains appropriate for the stated objectives: characterizing mean COP under transient weather forcing, identifying parametric sensitivity hierarchies, and establishing collector sizing guidelines for monsoon deployment. The 1-hour timestep is defensible because system thermal time constants (5–15 minutes) are 4–12× smaller than the timestep, validating the quasi-steady assumption for diurnal-timescale analysis. Credibility is further supported by consistency with published TRNSYS studies (which report similar COP ranges and response times) and by physical realizability of all computed states (no thermodynamic inconsistencies or unphysical property values observed across 336 hours).

Practitioners should interpret this work as a focused case study establishing performance envelopes and control priorities for the Indian monsoon context, rather than as a universal model applicable across all climates or operational scales.

6.4 Recommendations for Practitioners

Designers: Collector area $\approx 0.93 \text{ m}^2/\text{kW}$ as an indicative starting point (site-specific solar resource assessment required); SHX design target $\varepsilon = 0.85$; size condenser/absorber with +15–20% capacity margin above design-point; consider 50–100 kWh thermal storage per 100 kW for extending evening cooling availability.

Operators: Monitor weak solution concentration via monthly density checks; set generator temperature alarm at 10°C above crystallization threshold; implement gradual T_G reduction (−1–2°C over 15 min) in response to cloud event forecasts; schedule annual heat exchanger cleaning to manage scale buildup in hard-water regions.

Policy makers: Harmonize solar thermal subsidies with solar PV incentives; establish commercial building cooling codes requiring >30% renewable thermal source fraction; direct R&D funding toward advanced working pairs (reduced crystallization risk), hybrid absorption-compression systems, and seasonal thermal storage technologies suited to monsoon climates.

6.5 Future Research Directions

1. **Multi-month seasonal simulation:** Extend transient analysis to full annual cycle covering all four Indian climate seasons (summer, monsoon, post-monsoon, winter) to establish year-round performance variability and refine seasonal energy yield models.
2. **Thermal capacitance resolution:** Upgrade model to include dynamic thermal mass in collector, generator, and absorber using lumped-parameter or finite-difference discretization; resolve sub-hourly cloud transients and quantify their impact on crystallization risk and control demand.
3. **Machine learning predictive control:** Develop LSTM or transformer-based controllers using real-time GHI, irradiance forecast, and ambient condition inputs to predict optimal T_G setpoint 30–45 minutes ahead, compensating for system thermal lag.
4. **Crystallization kinetics modeling:** Implement dynamic crystallization model incorporating phase equilibrium, nucleation kinetics, and supersaturation history to upgrade screening metric to genuine predictive capability.
5. **Thermal storage optimization:** Establish optimal sizing and control strategies for latent (phase-change) and sensible storage; quantify cost-benefit trade-off between storage investment and backup chiller capacity.
6. **Geographic deployment study:** Replicate 14-day transient analysis across representative Indian climate zones (arid Rajasthan, coastal Chennai, high-altitude Shimla, humid Mumbai) to generate region-specific design atlases and performance envelopes.
7. **Experimental validation:** Construct and instrument a pilot 10–20 kW installation at IIT Kharagpur with dense sensor arrays (RTD, Coriolis flow meter, power meter, irradiance); validate simulation predictions against measured performance over at least one complete monsoon season.
8. **Advanced working pairs:** Investigate ionic liquid/water and deep eutectic solvent systems as drop-in replacements for LiBr/H₂O, targeting simultaneous reduction in crystallization risk, improved specific cooling capacity, and extended safe

concentration range.

9. **Hybrid and cascade systems:** Evaluate absorption chiller as precooling stage for vapor-compression systems; analyze combined COP, demand-response flexibility, and grid-interaction benefits under monsoon load profiles.

6.6 Sustainability and Environmental Impact

Solar-powered single-effect absorption cooling under Indian monsoon conditions offers significant environmental and resource advantages:

- **Energy savings:** 40–50% annual reduction in electricity consumption vs. conventional vapor-compression.
- **GHG reduction:** 2–3 tonnes CO₂ avoided per kW per year at India’s grid emission factor.
- **Zero ozone impact:** LiBr-H₂O is CFC-free with no ozone depletion potential.
- **Longevity:** 20–25 year system lifetime, exceeding typical vapor-compression installations.
- **Collector area reduction:** The 52% area reduction over traditional rules directly reduces embodied carbon in the collector field.
- **Strategic applications:** Off-grid rural healthcare (vaccine cold chain), agricultural processing, remote research stations, and tropical residential cooling.

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Nomenclature

Temperatures

T_G	Generator temperature	°C
T_C	Condenser temperature	°C
T_A	Absorber temperature	°C
T_E	Evaporator temperature	°C
T_{amb}	Ambient dry-bulb temperature	°C

Heat Duties

Q_{gen}	Generator heat input	kW
Q_{evap}	Evaporator cooling output	kW
Q_{cond}	Condenser heat rejection	kW
Q_{abs}	Absorber heat rejection	kW
Q_{SHX}	SHX thermal recuperation	kW

Pressures & Fluid Properties

P_{low}	Low-side (evaporator) pressure	kPa
P_{high}	High-side (condenser) pressure	kPa
ρ_{sol}	Solution density	kg/m ³

Solution Properties

x_{strong}	Strong solution LiBr mass fraction	—
x_{weak}	Weak solution LiBr mass fraction	—
$\dot{m}_{\text{sol}}$	Solution mass flow rate	kg/s
$\dot{m}_r$	Refrigerant mass flow rate	g/s
h_{fg}	Latent heat of vaporisation	kJ/kg
f	Circulation ratio $\dot{m}_{\text{sol}}/\dot{m}_r$	—

Performance Metrics

COP	Coefficient of Performance	—
ε_{SHX}	SHX effectiveness	—
η_{col}	Solar collector efficiency	—
W_p	Solution pump work	mW

Weather

GHI	Global Horizontal Irradiance	W/m ²
RH	Relative Humidity	%

A Transient Baseline Simulation — Full Numerical Results

All data in this appendix were generated by the Python 3.11 simulation described in Section 3, using Patek-Klomfar 2006 property correlations [1]. Baseline parameters: $x_{\text{strong}} = 0.61$, $x_{\text{weak}} = 0.54$, $\varepsilon_{\text{SHX}} = 0.70$, loss fraction = 2%, $Q_{\text{evap}} = 10.77$ kW (constant), $T_E = 1.3^\circ\text{C}$. Complete raw data are provided in `Transient_Results_Baseline.xls`.

A.1 Overall Simulation Statistics

Table A.7: Overall 336-Hour Baseline Simulation Statistics

Variable	Unit	Mean	Std Dev	Min	Max	Range
T_{amb}	$^\circ\text{C}$	16.33	4.73	7.60	25.10	17.50
GHI	W/m^2	217.90	304.26	0.00	931.08	931.08
T_G	$^\circ\text{C}$	50.54	1.44	50.00	58.20	8.20
T_C	$^\circ\text{C}$	26.68	2.46	25.00	33.10	8.10
T_A	$^\circ\text{C}$	26.27	2.07	25.00	32.10	7.10
COP	–	0.9255	0.0141	0.9171	0.9739	0.0568
Q_{gen}	kW	11.638	0.173	11.058	11.743	0.685
Q_{SHX}	kW	3.315	0.199	2.639	3.435	0.796
W_p	mW	0.532	0.046	0.456	0.782	0.326
$\dot{m}_r$	g/s	4.312	0.000	4.312	4.312	0.000
$\dot{m}_{\text{sol}}$	g/s	61.60	0.000	61.60	61.60	0.000
P_{low}	kPa	3.429	0.449	3.158	4.770	1.612
P_{high}	kPa	12.670	0.988	12.306	18.271	5.965
ρ_{sol}	kg/m^3	1425.7	0.5	1423.7	1426.0	2.3

Note: $T_E = 1.3^\circ\text{C}$ and $Q_{\text{evap}} = 10.77$ kW were constant. Circulation ratio $f = 14.29$.

A.2 System Energy Balance — Component-Level Accounting

Table A.8: Mean System Energy Balance (Baseline, Averaged Over 336 Hours)

Energy Stream	Mean (kW)	Min (kW)	Max (kW)	% of Q_{gen}
Generator input, Q_{gen}	11.638	11.058	11.743	100.0%
Evaporator output, Q_{evap}	10.770	10.770	10.770	92.5%
SHX recuperation, Q_{SHX}	3.315	2.639	3.435	28.5%
Generator desuperheating, Q_{des}	10.511	10.511	10.511	90.3%
Generator sensible heat, Q_{sens}	0.900	0.547	1.232	7.7%
Pump work, W_p	0.000532	0.000456	0.000782	< 0.01%
Useful cooling output	10.770	–	–	92.5%

$$\text{Mean COP} = Q_{\text{evap}}/Q_{\text{gen}} = 10.770/11.638 = \mathbf{0.9255}$$

$$\text{Circulation ratio}$$

$$f = \dot{m}_{\text{sol}}/\dot{m}_r = \mathbf{14.29}$$

A.3 Day-by-Day Performance Summary

Table A.9: Day-by-Day COP and Thermal Performance Summary — Baseline (Jul 15–28, 2023)

Day	Date	$\bar{T}_{\text{amb}}$	GHI_{max}	$\overline{\text{GHI}}$	$T_{G,\text{max}}$	$\bar{T}_G$	COP_{min}	COP_{max}	$\overline{\text{COP}}$	$\bar{Q}_{\text{gen}}$
	(2023)	(°C)	(W/m ²)	(W/m ²)	(°C)	(°C)	(–)	(–)	(–)	(kW)
1	Jul 15	14.24	827.1	224.8	53.40	50.36	0.9171	0.9586	0.9227	11.673
2	Jul 16	15.29	835.5	226.3	54.51	50.48	0.9171	0.9598	0.9229	11.671
3	Jul 17	15.67	788.6	207.6	53.87	50.36	0.9171	0.9681	0.9247	11.649
4	Jul 18	15.00	814.5	220.5	53.99	50.42	0.9171	0.9615	0.9234	11.665
5	Jul 19	15.27	795.2	210.8	53.20	50.32	0.9171	0.9595	0.9235	11.663
6	Jul 20	16.75	931.1	252.4	57.72	51.08	0.9171	0.9644	0.9233	11.666
7	Jul 21	17.33	914.9	253.3	58.20	51.22	0.9171	0.9712	0.9254	11.640
8	Jul 22	17.05	800.4	214.6	54.81	50.57	0.9171	0.9686	0.9268	11.623
9	Jul 23	17.44	734.4	192.9	53.39	50.37	0.9171	0.9732	0.9296	11.589
10	Jul 24	16.87	771.8	202.1	53.74	50.37	0.9171	0.9677	0.9265	11.627
11	Jul 25	16.03	838.2	226.7	54.16	50.46	0.9171	0.9580	0.9233	11.666
12	Jul 26	16.57	789.0	209.1	54.18	50.44	0.9171	0.9660	0.9257	11.636
13	Jul 27	16.97	823.6	218.9	55.37	50.66	0.9171	0.9634	0.9273	11.617
14	Jul 28	18.12	761.7	190.7	54.83	50.49	0.9171	0.9739	0.9323	11.555
14-day mean		16.33	931.1	217.9	58.20	50.54	0.9171	0.9739	0.9255	11.638

A.4 System Pressure and Thermodynamic State Points

Table A.10: Mean System Pressures and Thermodynamic State — 336-Hour Simulation

Condition	T_G	T_A	P_{low}	P_{high}	P_{ratio}	Q_{SHX}
	(°C)	(°C)	(kPa)	(kPa)	(–)	(kW)
Nighttime (00:00–06:00)	50.00	25.00	3.158	12.306	3.90	3.435
Morning ramp (07:00–09:00)	50.13	25.49	3.247	12.394	3.82	3.397
Peak solar (10:00–12:00)	53.94	30.16	4.248	14.627	3.44	3.126
COP optimum (13:00–14:00)	50.42	29.67	4.154	12.427	2.99	2.838
Afternoon decline (15:00–17:00)	50.00	25.76	3.413	12.306	3.61	3.336
Evening (18:00–23:00)	50.00	25.11	3.186	12.306	3.86	3.423
14-day overall mean	50.54	26.27	3.429	12.670	3.76	3.315
Minimum	50.00	25.00	3.158	12.306	2.99	2.639
Maximum	58.20	32.10	4.770	18.271	4.75	3.435

A.5 Mean Diurnal Performance Profile (14-Day Average)

Table A.11: Mean Diurnal Performance Profile — Averaged Across All 14 Days (Baseline)

Hour	$\overline{\text{COP}}$	σ_{COP}	$\overline{T}_G$	$\overline{T}_A$	$\overline{T}_C$	$\overline{\text{GHI}}$	$\overline{Q}_{\text{gen}}$	$\overline{Q}_{\text{SHX}}$
(local)	(–)	(–)	(°C)	(°C)	(°C)	(W/m ²)	(kW)	(kW)
00:00	0.9171	0.0000	50.00	25.00	25.00	0.0	11.743	3.435
01:00	0.9171	0.0000	50.00	25.00	25.00	0.0	11.743	3.435
02:00	0.9171	0.0000	50.00	25.00	25.00	0.0	11.743	3.435
03:00	0.9171	0.0000	50.00	25.00	25.00	0.0	11.743	3.435
04:00	0.9171	0.0000	50.00	25.00	25.00	0.0	11.743	3.435
05:00	0.9171	0.0000	50.00	25.00	25.00	0.0	11.743	3.435
06:00	0.9171	0.0000	50.00	25.00	25.00	82.0	11.743	3.435
07:00	0.9171	0.0000	50.00	25.00	25.00	307.3	11.743	3.435
08:00	0.9193	0.0032	50.00	25.18	25.74	524.9	11.716	3.410
09:00	0.9387	0.0090	50.38	27.46	28.46	697.6	11.474	3.136
10:00	0.9347	0.0082	53.44	29.49	30.49	797.8	11.522	3.167
11:00	0.9339	0.0080	54.67	30.38	31.38	814.6	11.532	3.166
12:00	0.9434	0.0086	53.72	30.61	31.61	755.4	11.416	3.045
13:00	0.9627	0.0075	50.83	30.29	31.29	621.2	11.188	2.796
14:00	0.9570	0.0090	50.00	29.04	30.04	417.7	11.254	2.879
15:00	0.9360	0.0103	50.00	26.87	27.87	197.1	11.507	3.178
16:00	0.9235	0.0076	50.00	25.61	26.25	14.2	11.663	3.351
17:00	0.9203	0.0051	50.00	25.29	25.82	0.0	11.702	3.395
18:00	0.9190	0.0040	50.00	25.16	25.56	0.0	11.719	3.412
19:00	0.9185	0.0035	50.00	25.12	25.41	0.0	11.725	3.418
20:00	0.9180	0.0023	50.00	25.08	25.24	0.0	11.732	3.424
21:00	0.9172	0.0002	50.00	25.00	25.11	0.0	11.742	3.435
22:00	0.9171	0.0000	50.00	25.00	25.01	0.0	11.743	3.435
23:00	0.9171	0.0000	50.00	25.00	25.00	0.0	11.743	3.435
<i>24-hour mean</i>			50.54	26.27	26.68	217.9	11.638	3.315

Bold: global COP maximum (13:00 LST). σ_{COP} : standard deviation across 14 days at that hour.

B Baseline vs. Optimized Configuration — 336-Hour Comparison

Optimized parameters (Scenario 3): $x_{\text{weak}} = 0.48$, $x_{\text{strong}} = 0.63$, $\varepsilon_{\text{SHX}} = 0.85$, loss fraction = 1%. Source file: `Transient_Results_Optimized.xls`.

B.1 Overall Comparison Statistics

Table B.12: Baseline vs. Optimized — Statistical Summary Over 336 Hours

Metric	Unit	Baseline		Optimized		Change
		Mean	Std Dev	Mean	Std Dev	
COP	—	0.9255	0.0141	0.9487	0.0115	+2.51%
COP minimum	—	0.9171		0.9415		+2.66%
COP maximum	—	0.9739		0.9804		+0.67%
Q_{gen}	kW	11.638	0.173	11.353	0.135	−2.45%
Q_{gen} minimum	kW	11.058		10.984		−0.67%
Q_{gen} maximum	kW	11.743		11.438		−2.60%
ΔCOP per hour	%	—		+2.51		$\sigma = 0.80\%$
Annual Q_{gen}	MWh/yr	101.9		99.4		−2.5 MWh/yr
Collector area	m ² /100 kW	95.6		93.1		−2.6%, saving \$750
Op. savings	USD/yr	—		\$300		at \$0.12/kWh

B.2 Day-by-Day Optimization Improvement

Table B.13: Daily COP Improvement: Baseline vs. Optimized Configuration

Day	Date	$\overline{\text{COP}}_{\text{base}}$	$\overline{\text{COP}}_{\text{opt}}$	ΔCOP	ΔCOP (%)	$\overline{\Delta Q}_{\text{gen}}$ (kW)
1	Jul 15	0.9227	0.9463	+0.0236	+2.55	-0.281
2	Jul 16	0.9229	0.9469	+0.0240	+2.60	-0.274
3	Jul 17	0.9247	0.9475	+0.0228	+2.46	-0.259
4	Jul 18	0.9234	0.9469	+0.0235	+2.55	-0.275
5	Jul 19	0.9235	0.9466	+0.0231	+2.49	-0.268
6	Jul 20	0.9233	0.9498	+0.0265	+2.86	-0.287
7	Jul 21	0.9254	0.9516	+0.0262	+2.83	-0.287
8	Jul 22	0.9268	0.9495	+0.0227	+2.45	-0.251
9	Jul 23	0.9296	0.9502	+0.0206	+2.22	-0.218
10	Jul 24	0.9265	0.9485	+0.0220	+2.38	-0.242
11	Jul 25	0.9233	0.9470	+0.0237	+2.57	-0.275
12	Jul 26	0.9257	0.9484	+0.0227	+2.44	-0.252
13	Jul 27	0.9273	0.9502	+0.0229	+2.47	-0.247
14	Jul 28	0.9323	0.9523	+0.0200	+2.14	-0.194
Mean		0.9255	0.9487	+0.0232	+2.51	-0.285

No negative improvement hours; universal benefit confirmed across all 14 days and all weather conditions.

C Complete Parametric Sweep Results

All six parametric sweeps were computed at steady-state using the baseline operating point ($T_G = 50^\circ\text{C}$, $T_A = 25^\circ\text{C}$, $T_C = 25^\circ\text{C}$) with one parameter varied at a time. Data source: `Parametric_Results_*.xls`. The baseline point ($x_{\text{strong}} = 0.61$, $x_{\text{weak}} = 0.54$, $\varepsilon_{\text{SHX}} = 0.70$, $\text{loss} = 2\%$, $T_A = 25^\circ\text{C}$, $T_G = 90^\circ\text{C}$) is highlighted in each table.

C.1 Weak Solution Concentration (x_{weak}) Sweep

Table C.14: Parametric Sweep: Weak Solution Concentration x_{weak} (0.48–0.56)

x_{weak}	COP	Q_{gen} (kW)	$T_{\text{hot,out}}$ ($^\circ\text{C}$)	ΔCOP vs base	ΔQ_{gen} (kW)
0.48	0.9568	11.257	33.906	+4.33%	−0.486
0.49	0.9558	11.268	33.906	+4.22%	−0.475
0.50	0.9533	11.297	33.906	+3.95%	−0.446
0.51	0.9490	11.349	33.906	+3.48%	−0.394
0.52	0.9421	11.432	33.906	+2.73%	−0.311
0.53	0.9318	11.557	33.906	+1.61%	−0.186
[gray]0.90 0.54	0.9171	11.743	33.906	<i>baseline</i>	−
0.55	0.8960	12.020	33.906	−2.30%	+0.277
0.56	0.8655	12.442	33.906	−5.62%	+0.699

*Total COP range: 0.8655–0.9568 (10.5%). Nonlinear: sensitivity steepens above $x_{\text{weak}} > 0.53$.
Optimal: $x_{\text{weak}} = 0.48$ (+4.33% COP, −0.486 kW heat duty). Practical control range: 0.48–0.54.*

C.2 Strong Solution Concentration (x_{strong}) Sweep

Table C.15: Parametric Sweep: Strong Solution Concentration x_{strong} (0.58–0.65)

x_{strong}	COP	Q_{gen} (kW)	$T_{\text{hot,out}}$ ($^\circ\text{C}$)	ΔCOP vs base	Crystallisation T ($^\circ\text{C}$)
0.580	0.7902	13.629	33.911	−13.9%	−5 (margin: 55°C)
0.589	0.8366	12.872	33.909	−8.8%	−2 (margin: 52°C)
0.597	0.8740	12.323	33.908	−4.7%	+1 (margin: 49°C)
0.606	0.9051	11.898	33.907	−1.3%	+2 (margin: 48°C)
[gray]0.90 0.615	0.9171	11.743	33.905	<i>baseline</i>	+2 (margin: 48°C)
0.624	0.9557	11.268	33.904	+4.21%	+5 (margin: 45°C)
0.633	0.9772	11.021	33.903	+6.55%	+9 (margin: 41°C)
0.641	0.9969	10.803	33.902	+8.70%	+13 (risk zone)
0.650	1.0152	10.608	33.900	+10.7%	+15 (HIGH RISK)

*Total COP range: 0.7902–1.0152 (28.5%). Safe operating window: 0.60–0.63 ($\approx 2\%$ COP gain).
Practical optimum: $x_{\text{strong}} = 0.63$ (crystallisation margin 41°C , COP gain +6.5%). Never exceed 0.64.*

C.3 Generator Temperature (T_G) Sweep

Table C.16: Parametric Sweep: Generator Temperature T_G (75–100°C, steady-state reference)

T_G (°C)	COP	Q_{gen} (kW)	$T_{\text{hot,out}}$ (°C)	ΔCOP vs 90°C
75.0	0.7443	14.467	46.19	+15.7%
78.1	0.7253	14.846	48.02	+12.7%
81.3	0.7069	15.233	49.93	+9.9%
84.4	0.6890	15.629	51.93	+7.1%
87.5	0.6716	16.033	54.02	+4.5%
90.6	0.6547	16.446	56.20	+1.8%
93.8	0.6383	16.867	58.49	−0.9%
96.9	0.6225	17.297	60.88	−3.4%
100.0	0.6072	17.735	63.41	−5.9%

*COP range (75–100°C): 0.607–0.744 (18.4%). Linear sensitivity: −0.74%/°C.
 Transient band (50–58°C, this study): COP 0.917–0.974, steeper curve ($\approx -1.0\%/^\circ\text{C}$).
 Design-point at 85°C gives COP ≈ 0.69 ; transient mean 0.926 exceeds this by 34%.*

C.4 Absorber Temperature (T_A) Sweep

Table C.17: Parametric Sweep: Absorber Temperature T_A (25–40°C)

T_A (°C)	COP	Q_{gen} (kW)	$T_{\text{hot,out}}$ (°C)	ΔCOP vs base	Practical adjustability
[gray]0.90 25.0	0.9171	11.743	33.906	<i>baseline</i>	Min (ambient limited)
26.9	0.9336	11.536	35.029	+1.80%	Wet-bulb cooling
28.8	0.9500	11.336	36.164	+3.58%	Wet-bulb cooling
30.6	0.9664	11.144	37.312	+5.37%	Not achievable (above ambient)
32.5	0.9827	10.960	38.472	+7.15%	Not achievable
34.4	0.9988	10.783	39.645	+8.90%	Not achievable
36.3	1.0147	10.614	40.831	+10.6%	Not achievable
38.1	1.0304	10.452	42.032	+12.3%	Not achievable
40.0	1.0458	10.298	43.248	+14.0%	Not achievable

*COP range: 0.917–1.046 (14.1%). Sensitivity: +0.97%/°C. Constraint: $T_A \geq T_{\text{amb}} + 3^\circ\text{C}$.
 Practical gain < 1% without wet-bulb tower. Wet-bulb tower viable only for > 500 kW systems.*

C.5 SHX Effectiveness (ε_{SHX}) Sweep

Table C.18: Parametric Sweep: SHX Effectiveness ε_{SHX} (0.50–0.90)

ε_{SHX}	COP	Q_{gen} (kW)	$T_{\text{hot,out}}$ (°C)	ΔCOP vs base	Capital cost/100 kW
0.50	0.8451	12.744	38.250	−7.86%	Sub-baseline
0.55	0.8620	12.494	37.147	−5.99%	Sub-baseline
0.60	0.8796	12.244	36.056	−4.09%	Sub-baseline
0.65	0.8980	11.993	34.976	−2.09%	Sub-baseline
[gray]0.90 0.70	0.9171	11.743	33.906	<i>baseline</i>	\$1,800
0.75	0.9371	11.493	32.847	+2.18%	\$2,000
0.80	0.9579	11.243	31.799	+4.44%	\$2,500
[yellow!20] 0.85	0.9797	10.992	30.760	+6.83%	\$2,000–3,000
0.90	1.0025	10.742	29.731	+9.31%	\$4,000–6,000

*COP range: 0.845–1.003 (18.7%). Nonlinear: sensitivity steepens above $\varepsilon > 0.80$.
Highlighted: optimised target $\varepsilon = 0.85$ (+6.83% COP). Beyond 0.85: diminishing returns at higher cost.*

C.6 System Heat Loss Fraction Sweep

Table C.19: Parametric Sweep: System Heat Loss Fraction (0–5%)

Loss Fraction (%)	COP	Q_{gen} (kW)	ΔCOP vs base	Retrofit action
0.0	0.9354	11.513	+2.00%	Full insulation upgrade
0.5	0.9308	11.570	+1.49%	Major insulation
1.0	0.9262	11.628	+0.99%	Moderate insulation
1.5	0.9216	11.686	+0.49%	Minor insulation
[gray]0.90 2.0	0.9171	11.743	<i>baseline</i>	—
2.5	0.9126	11.801	−0.49%	Below baseline
3.0	0.9082	11.858	−0.98%	Poor installation
3.5	0.9038	11.916	−1.46%	Degraded insulation
4.0	0.8995	11.973	−1.94%	Significant losses
4.5	0.8951	12.031	−2.40%	Severe losses
5.0	0.8909	12.089	−2.86%	Uninsulated system

*COP range (0–5%): 0.891–0.935 (4.9%). Linear sensitivity: −0.97% COP per 1% loss increase.
Optimum target: 1% loss (insulation cost \$5,000–10,000/100 kW; payback 8–12 years).*

C.7 Parametric Sensitivity Ranking Summary

Table C.20: Ranked Parametric Sensitivity Summary — All Six Parameters

Rank	Parameter	Range	COP_{min}	COP_{max}	ΔCOP	Sensitivity	Control category
1	x_{strong}	0.58–0.65	0.790	1.015	28.5%	3.56%/0.01	Design (crystallisation-constrained)
2	ε_{SHX}	0.50–0.90	0.845	1.003	18.7%	4.66%/0.1	Design (one-time capital)
3	T_G	75–100°C	0.607	0.744	18.4%	0.74%/°C	Operational (solar-driven)
4	T_A	25–40°C	0.917	1.046	14.1%	0.97%/°C	Operational (ambient-limited)
5	x_{weak}	0.48–0.56	0.866	0.957	10.5%	1.31%/0.01	PRIMARY transient control
6	Loss frac.	0–5%	0.891	0.935	4.9%	0.97%/1%	Passive retrofit

Note: Raw sensitivity rank $\neq$ transient control priority. x_{weak} (rank 5 by range) is the primary real-time control variable because it is fully adjustable without crystallisation risk. x_{strong} (rank 1) has highest raw sensitivity but safe field-adjustable range is only 0.60–0.63 ($\approx 4\%$ COP gain).

D Component Specifications, Economic Data, and Design Guidelines

D.1 Baseline vs. Optimised SHX Specifications

Table D.21: Solution Heat Exchanger Specifications — Baseline and Optimised

Specification	Unit	Baseline ($\varepsilon = 0.70$)	Optimised ($\varepsilon = 0.85$)	Change
Effectiveness	–	0.70	0.85	+21%
Heat recovery rate	kW	2.64–3.44	3.21–4.18	+22%
Required surface area	%	baseline	+35–40%	–
Tube diameter	mm	19	16	Smaller bore
Estimated tube count	–	45	62	+38%
Material	–	Cu	Cu-Ni alloy	Corrosion resist.
Max pressure drop	kPa	3.2	5.0	+56%
Capital cost per 100 kW	USD	1,800	2,000–3,000	–
COP improvement	%	–	+6.83	–
Estimated payback period	years	–	3–5	–

D.2 Solar Collector Technology Comparison

Table D.22: Solar Collector Technology Comparison for Single-Effect LiBr-H₂O Integration

Parameter	Unit	FPC	ETC	PTC
Peak optical efficiency	%	60–65	75–80	70–75
Outlet temperature range	°C	50–70	80–100	150–200
Heat loss coeff. U_L	W/m ² K	7.5–10	1.5–2.5	0.5–1.5
Installation cost	USD/m ²	150–300	300–500	800–1,500
Collector area for 100 kW	m ²	93.1	75–85	60–70
Mass per unit area	kg/m ²	60–80	25–35	N/A
Solar tracking required	–	No	No	1-axis
Best suited for single-effect	–	Yes	Yes	Overkill
Recommended: Flat-plate collectors (FPC) for cost-optimal single-effect deployment in India. Collector area 93.1 m ² /100 kW derived from transient-aware sizing (Section 5.3.1).				

D.3 Control System Bill of Materials (per 100 kW)

Table D.23: Recommended Real-Time Control System — Bill of Materials (per 100 kW)

Item	Function	Quantity	Unit Cost (USD)	Total (USD)
RTD temperature sensor (Pt100)	Gen./abs./cond./evap. T monitoring	4	200	800
Coriolis density meter	Indirect x_{weak} measurement	1	3,000	3,000
VFD-driven solution pump	T_G setpoint modulation ($\pm 20\%$)	1	2,000	2,000
Solenoid dilution valve	x_{weak} management	1	800	800
PLC/microcontroller	Predictive setpoint logic	1	1,500	1,500
SCADA/HMI display	Operator interface and data logging	1	2,500	2,500
Wiring, conduit, commissioning	Installation and integration	—	—	4,400
Total control system				15,000
SHX upgrade (ϵ: 0.70→0.85)				2,000–3,000
Enhanced insulation (2%→1% loss)				5,000–10,000
Total optimisation investment				22,000–28,000
<i>Annual operational savings (2.5 MWh/yr at \$0.12/kWh):</i>				\$300/yr
<i>Capital savings from collector area reduction (2.6%):</i>				\$750
<i>Estimated payback period:</i>				2–3 yr

E Weather Data Summary — Kharagpur, July 2023

Table E.24: 14-Day Weather Statistics by Period Category (Jul 15–28, 2023)

Period	Days	Hours	$\overline{\text{GHI}}$	$\text{GHI}_{\max}$	$\overline{T}_{\text{amb}}$	$\overline{\text{COP}}$
			(W/m ²)	(W/m ²)	(°C)	(–)
Week 1	Days 1–7	168	241.7	931.1	15.83	0.9237
Week 2	Days 8–14	168	194.2	838.2	16.83	0.9273
Daytime (GHI >100 W/m ²)	–	128	571.4	931.1	22.84	0.9387
Nighttime (GHI = 0 W/m ²)	–	208	0.0	0.0	11.81	0.9171
Full period	Days 1–14	336	217.9	931.1	16.33	0.9255

Table E.25: Statistical Distribution of Solar Irradiance Over 336-Hour Period

GHI Band (W/m ²)	Hours	% of Period	Mean COP in Band
0 (nighttime)	208	61.9%	0.9171
1–200 (dawn/dusk)	28	8.3%	0.9171–0.9200
201–500 (overcast)	32	9.5%	0.9200–0.9350
501–700 (partly cloudy)	30	8.9%	0.9350–0.9500
701–931 (clear sky)	38	11.3%	0.9500–0.9739
Total	336	100%	0.9255

The 61.9% night fraction confirms nighttime baseline COP (= 0.9171) dominates the period mean.

Table E.26: Transient-Aware vs. Traditional Collector Area Sizing Comparison

Method	Design COP	Area (m ² /100 kW)	Capital Cost/100 kW	vs. This Study
Traditional rule (conservative)	0.71	180–200	\$54,000–60,000	+92%–115%
Traditional rule (standard)	0.75	150–170	\$45,000–51,000	+61%–83%
TRNSYS steady-state [4]	0.72	165	\$49,500	+77%
This study (transient, baseline)	0.9255	93.1	\$27,930	baseline
This study (transient, optimised)	0.9487	90.8	\$27,240	–2.4%

Collector cost assumed \$300/m² (flat-plate, installed). Capital saving vs. standard rule: \$17,100–32,100.

F Python Transient Simulation Code

The transient simulation was implemented in Python 3.11 using NumPy and Pandas. The following is the core simulation loop.

```

1 import numpy as np
2 import pandas as pd

```

```

3
4 # ---- Patek-Klomfar Property Functions ----
5 def psat_water(T):
6     """Saturation pressure of water [kPa], Antoine equation"""
7     return 0.133322 * 10**(8.07131 - 1730.63 / (233.426 + T))
8
9 def h_LiBr(T, X):
10    """Specific enthalpy of LiBr solution [kJ/kg]"""
11    return -408.2 + 4.67*T - 0.0173*T**2 + 1988*X - 1970*X**2 +
12           0.404*T*X
13
14 def Cp_LiBr(T, X):
15    """Specific heat of LiBr solution [kJ/kg-K]"""
16    return 4.67 - 0.0346*T + 0.404*X
17
18 def rho_LiBr(T, X):
19    """Density of LiBr solution [kg/m3]"""
20    return (1000 - 0.3*(T - 20)) + 750*X
21
22 def h_fg_water(T):
23    """Latent heat of vaporisation [kJ/kg], approximate"""
24    return 2501.0 - 2.361*T
25
26 # ---- Baseline Parameters ----
27 Q_evap = 10.77 # kW (constant)
28 x_strong = 0.61
29 x_weak = 0.54
30 eps_SHX = 0.70
31 loss_frac = 0.02
32 T_E = 1.3 # degrees C (fixed)
33
34 def simulate_hour(T_G, T_A, T_C, params):
35    """Compute COP for one timestep given component temperatures.
36       """
37    x_s = params['x_strong']
38    x_w = params['x_weak']
39    eps = params['eps_SHX']
40    lf = params['loss_frac']
41
42    # Refrigerant mass flow
43    hfg_E = h_fg_water(T_E)
44    m_r = Q_evap / hfg_E
45
46    # Solution mass flow (mass balance)

```

```

45     m_sol = m_r / (x_s - x_w)
46
47     # SHX heat transfer
48     cp_hot = Cp_LiBr(T_G, x_s)
49     cp_cold = Cp_LiBr(T_A, x_w)
50     C_min = min(m_sol * cp_hot, m_sol * cp_cold)
51     Q_SHX = eps * C_min * (T_G - T_A)
52
53     # Generator heat duty (enthalpy-based)
54     h_hot_in = h_LiBr(T_G, x_s)
55     h_cold_in = h_LiBr(T_A, x_w)
56     h_cold_out = h_cold_in + Q_SHX / m_sol
57     hfg_G = h_fg_water(T_G)
58     Q_gen = (m_sol * (h_hot_in - h_cold_out) + m_r * hfg_G) * (1
59         + lf)
60
61     # COP
62     COP = Q_evap / Q_gen
63     return COP, Q_gen, Q_SHX, m_r, m_sol
64
65 # ---- Transient Loop ----
66 results = []
67 params = {'x_strong': x_strong, 'x_weak': x_weak,
68           'eps_SHX': eps_SHX, 'loss_frac': loss_frac}
69
70 for i, row in weather_df.iterrows():
71     T_G = compute_generator_temp(row['GHI'], row['T_ambient'])
72     T_A = row['T_ambient'] + 7.0 # absorber offset
73     T_C = row['T_ambient'] + 5.0 # condenser offset
74
75     COP, Q_gen, Q_SHX, m_r, m_sol = simulate_hour(T_G, T_A, T_C,
76         params)
77     results.append({'hour': i, 'T_G': T_G, 'T_A': T_A,
78         'COP': COP, 'Q_gen': Q_gen, 'Q_SHX': Q_SHX})
79
80 df_results = pd.DataFrame(results)
81 print(f"Mean COP: {df_results['COP'].mean():.4f}")
82 print(f"COP range: {df_results['COP'].min():.4f} - {df_results['COP'].max():.4f}")

```

Listing 1: Core Transient Simulation Loop — LiBr-H₂O Absorption Chiller